# ctz-euclide

Typst Port

*Euclidean Geometry for Typst*

---

A comprehensive geometry package built on CeTZ

Version 0.1.5

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# 1. Introduction

`ctz-euclide` is a geometry package for Typst, a port of the LaTeX package `tkz-euclide`. Built on top of CeTZ (a powerful drawing library), it provides high-level constructions for Euclidean geometry.

## 1.1. Features

- **Point Registry**: Define points once, reference them by name throughout your figure
- **Geometric Constructions**: Perpendiculars, parallels, bisectors, mediators
- **Intersections**: Line–line, line–circle, circle–circle with multiple solution handling
- **Triangle Centers**: Centroid, circumcenter, incenter, orthocenter, and 10+ specialized centers
- **Special Triangles**: Medial, orthic, intouch triangles
- **Transformations**: Rotation, reflection, translation, homothety, projection, inversion
- **Drawing & Styling**: Points, labels, angles, segments with tick marks
- **Grid & Axes**: Coordinate systems with customizable appearance
- **Clipping**: Mathematical line clipping for clean bounded figures

## 1.2. Installation

Import the package in your Typst document:

```
#import "@preview/ctz-euclide:0.2.1": *
```

All figures use the `ctz-canvas` function (re-exported from CeTZ):

```
#ctz-canvas({
  import cetz.draw: *
  ctz-init()

  // Your geometry code here
})
```

Naming notes:

- All public functions are prefixed with `ctz-` to avoid conflicts.
- Point creation and drawing use `ctz-def-points` and `ctz-draw-points`.
- Other constructors use `ctz-def-*`, and drawing utilities use `ctz-draw-*`.

The `ctz-init()` call initializes the point registry and coordinate resolver.

## 1.3. Basic Usage

**Code**

```
#ctz-canvas(length: 0.8cm, {
  import cetz.draw: *
  ctz-init()

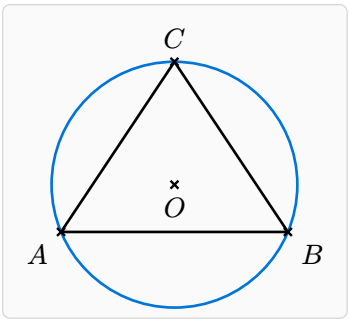
  // Define points
  ctz-def-points(A: (0, 0), B: (4, 0), C: (2, 3))

  // Draw triangle
  ctz-draw(line: ("A", "B", "C", "A"), stroke: black)

  // Find circumcenter and draw circumcircle
  ctz-def-circumcenter("O", "A", "B", "C")
  ctz-draw(circle-through: ("O", "A"), stroke: blue)

  // Draw and label points
  ctz-draw(points: ("A", "B", "C", "O"), labels: (
    A: "below left", B: "below right",
    C: "above", O: "below"))
})
```

**Figure**



# 2. Core Concepts

## 2.1. The Point Registry

The point registry is the heart of `ctz-euclide`. Once you define a point with a name, that name can be used directly in CeTZ drawing commands.

```
ctz-def-points(A: (0, 0), B: (3, 4))  // Register points A and B
ctz-draw(segment: ("A", "B"))              // Use them directly in CeTZ
```

Under the hood, `ctz-init()` installs a coordinate resolver that translates `"A"` to the stored coordinates.

## 2.2. Figure Scaling

Control the size of your figures using CeTZ's `length` parameter:

```
#ctz-canvas(length: 0.8cm, { ... })
```

This scales everything proportionally, including stroke widths. Typical values:

- `0.6cm` – small inline figures
- `0.8cm` – standard examples
- `1.0cm` – large detailed figures

## 2.3. Coordinate Systems

Points can be defined in multiple ways:

```
// Explicit coordinates
ctz-def-points(A: (2, 3))

// Using existing CeTZ coordinates
ctz-def-points(B: (rel: (1, 1), to: "A"))

// Mixed: numbers and existing points
ctz-def-points(C: (4, 0), D: "A", E: (3, 2))
```

# 3. Point Definitions

## 3.1. Basic Points — `ctz-def-points`

Define one or more points at specific coordinates:

```
ctz-def-points(A: (0, 0), B: (4, 0), C: (2, 3))
```

## 3.2. Midpoint — `ctz-def-midpoint`

Find the midpoint of a segment:

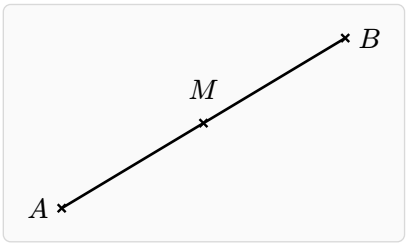
**Code**

```
#ctz-canvas(length: 0.8cm, {
  import cetz.draw: *
  ctz-init()

  ctz-def-points(A: (0, 0), B: (5, 3))
  ctz-def-midpoint("M", "A", "B")

  ctz-draw(segment: ("A", "B"), stroke: black)
  ctz-draw(points: ("A", "B", "M"), labels: (
    A: "left", B: "right", M: "above"))
})
```

**Figure**




## 3.3. Regular Polygons — `ctz-def-regular-polygon`

Generate vertices of a regular $n$-gon. If you pass a polygon name first, it is registered and can be drawn/labeled by name: You can also mark all sides during drawing with `mark` and optional `mark-opts`.

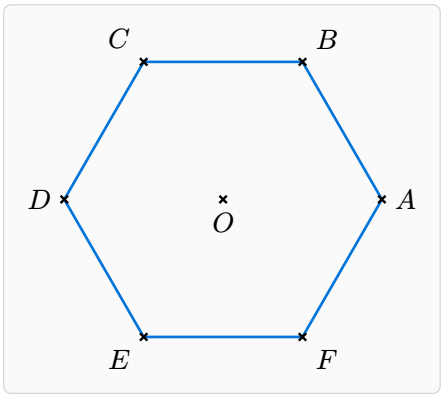
**Code**

```
#ctz-canvas(length: 0.7cm, {
  import cetz.draw: *
  ctz-init()

  ctz-def-points(O: (0, 0), A: (3, 0))
  // O is the center; A is the starting vertex that
fixes the radius/angle.
  ctz-def-regular-polygon("Hex", ("A", "B", "C", "D",
"E", "F"), "O", "A")

  ctz-draw("Hex", stroke: blue)
  ctz-draw(points: ("A", "B", "C", "D", "E", "F",
"O"), labels: (O: "below"))
})
```

**Figure**




## 3.4. Named Polygons — `ctz-def-polygon` / `ctz-label-polygon`

Define a polygon once and draw/label it by name:

```
ctz-def-points(A: (0, 0), B: (4, 0), C: (4, 2), D: (0, 2))
ctz-def-polygon("P1", "A", "B", "C", "D")
ctz-draw("P1", stroke: black)
ctz-label-polygon("P1", $P_1$, pos: "center")
```

## 3.5. Linear Combination — `ctz-def-linear`

Define a point along a line: $P = A + k(B - A)$

```
ctz-def-linear("P", "A", "B", 0.3)  // P is 30% from A to B
ctz-def-linear("Q", "A", "B", 1.5)  // Q extends beyond B
```

# 4. Line Constructions

## 4.1. Perpendicular — `ctz-def-perp`

Construct a perpendicular line through a point:

**Code**

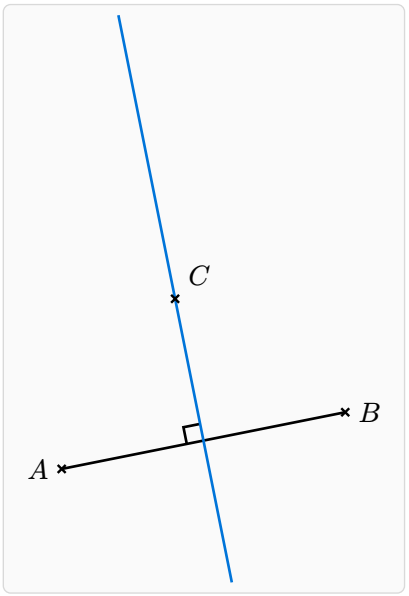
```
#ctz-canvas(length: 0.75cm, clip-canvas: (-0.5, -0.5,
5.5, 3.5), {
  import cetz.draw: *
  ctz-init()

  ctz-def-points(A: (0, 0), B: (5, 1), C: (2, 3))
  ctz-def-perp("P1", "P2", ("A", "B"), "C")
  ctz-def-project("H", "C", "A", "B")

  ctz-draw(segment: ("A", "B"), stroke: black)
  ctz-draw(segment: ("P1", "P2"), stroke: blue)
  ctz-draw-mark-right-angle("A", "H", "C", size: 0.3)

  ctz-draw(points: ("A", "B", "C"), labels: (
    A: "left", B: "right", C: "above"))
})
```

**Figure**




## 4.2. Parallel — `ctz-def-para`

Construct a parallel line through a point:

**Code**

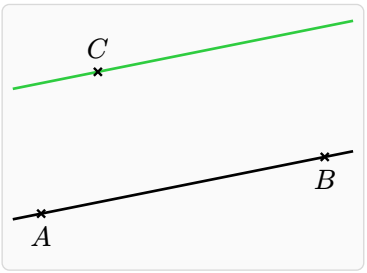
```
#ctz-canvas(length: 0.75cm, clip-canvas: (-0.5, -0.5,
5.5, 3.5), {
  import cetz.draw: *
  ctz-init()

  ctz-def-points(A: (0, 0), B: (5, 1), C: (1, 2.5))
  ctz-def-para("P1", "P2", ("A", "B"), "C")

  ctz-draw-line-add("A", "B", add: (2, 2), stroke:
black)
  ctz-draw-line-add("P1", "P2", add: (2, 2), stroke:
green)

  ctz-draw(points: ("A", "B", "C"), labels: (
    A: "below",
    B: "below",
    C: "above"
  ))
})
```

**Figure**




## 4.3. Angle Bisector — `ctz-def-bisect`

Construct the bisector of an angle:

**Code**

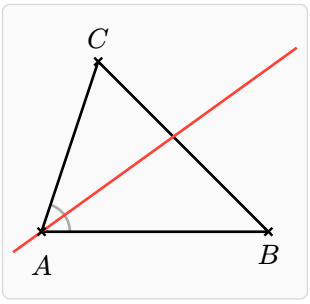
```
#ctz-canvas(length: 0.8cm, clip-canvas: (-0.5, -0.5,
4.5, 3.5), {
  import cetz.draw: *
  ctz-init()

  ctz-def-points(A: (0, 0), B: (4, 0), C: (1, 3))
  ctz-def-bisect("D1", "D2", "C", "A", "B")

  ctz-draw(line: ("A", "B", "C", "A"), stroke: black)
  ctz-draw-segment("D1", "D2", stroke: red)

  ctz-draw-angle("A", "C", "B", radius: 0.5, stroke:
gray)
  ctz-draw(points: ("A", "B", "C"), labels: (
    A: "below",
    B: "below",
    C: "above"
  ))
})
```

**Figure**




## 4.4. Perpendicular Bisector — `ctz-def-mediator`

Construct the perpendicular bisector of a segment:

**Code**

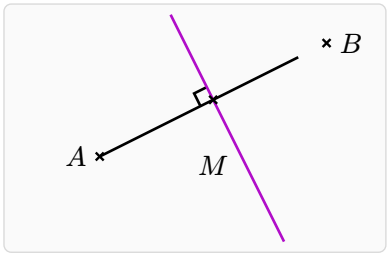
```
#ctz-canvas(length: 0.75cm, clip-canvas: (-0.5, -0.5,
4.5, 3.5), {
  import cetz.draw: *
  ctz-init()

  ctz-def-points(A: (1, 1), B: (5, 3))
  ctz-def-mediator("M1", "M2", "A", "B")
  ctz-def-midpoint("M", "A", "B")

  ctz-draw(segment: ("A", "B"), stroke: black)
  ctz-draw(segment: ("M1", "M2"), stroke: purple)
  ctz-draw-mark-right-angle("M1", "M", "A", size:
0.25)

  ctz-draw(points: ("A", "B", "M"), labels: (
    A: "left",
    B: "right",
    M: "below"
  ))
})
```

**Figure**



# 5. Intersections

## 5.1. Line–Line — `ctz-def-ll`

Find the intersection of two lines:

**Code**

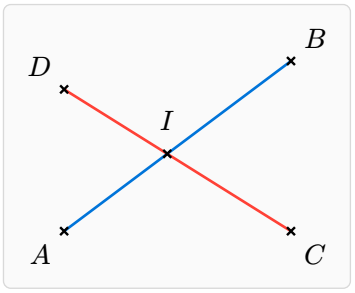
```
#ctz-canvas(length: 0.8cm, {
  import cetz.draw: *
  ctz-init()

  ctz-def-points(A: (0, 0), B: (4, 3),
            C: (4, 0), D: (0, 2.5))
  ctz-def-line("L1", "A", "B")
  ctz-def-line("L2", "C", "D")
  ctz-def-ll("I", "L1", "L2")

  ctz-draw("L1", stroke: blue)
  ctz-draw("L2", stroke: red)

  ctz-draw(points: ("A", "B", "C", "D", "I"), labels:
(I: "above"))
})
```

**Figure**




## 5.2. Line–Circle — `ctz-def-lc`

Find intersections of a line with a circle:

**Code**

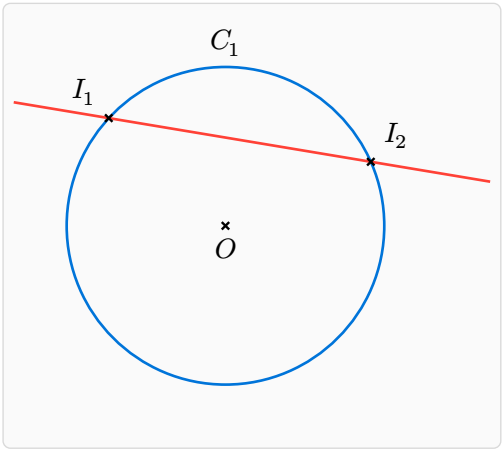
```
#ctz-canvas(length: 0.7cm, clip-canvas: (-4, -4, 5,
4), {
  import cetz.draw: *
  ctz-init()

  ctz-def-points(O: (0, 0), R: (3, 0),
            A: (-2, 2), B: (4, 1))
  ctz-def-line("L1", "A", "B")
  ctz-def-circle("C1", "O", through: "R")
  ctz-def-lc(("I1", "I2"), "L1", "C1")

  ctz-draw("C1", stroke: blue)
  ctz-label-circle("C1", $C_1$, pos: "above", dist:
0.2)
  ctz-draw-line-add("A", "B", add: (2, 2), stroke:
red)

  ctz-draw(points: ("O", "I1", "I2"), labels: (
    O: "below",
    I1: "above left",
    I2: "above right"
  ))
})
```

**Figure**




Named line/circle form:

```
ctz-def-line("L1", "A", "B")
ctz-def-circle("C1", "O", radius: 3)
ctz-def-lc(("I1", "I2"), "L1", "C1")
```

## 5.3. Circle–Circle — `ctz-def-cc`

Find intersections of two circles:

**Code**

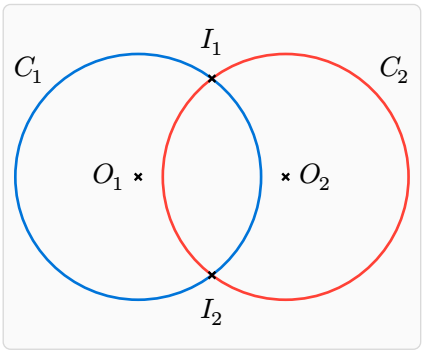
```
#ctz-canvas(length: 0.65cm, {
  import cetz.draw: *
  ctz-init()

  ctz-def-points(O1: (0, 0), O2: (3, 0),
            R1: (2.5, 0), R2: (5.5, 0))
  ctz-def-circle("C1", "O1", through: "R1")
  ctz-def-circle("C2", "O2", through: "R2")
  ctz-def-cc(("I1", "I2"), "C1", "C2")

  ctz-draw("C1", stroke: blue)
  ctz-draw("C2", stroke: red)
  ctz-label-circle("C1", $C_1$, pos: "above left",
dist: 0.2)
  ctz-label-circle("C2", $C_2$, pos: "above right",
dist: 0.2)

  ctz-draw(points: ("O1", "O2", "I1", "I2"), labels:
(
    I1: "above",
    I2: "below"
  ))
})
```

**Figure**




Named circle form:

```
ctz-def-circle("C1", "O1", through: "R1")
ctz-def-circle("C2", "O2", through: "R2")
ctz-def-cc(("I1", "I2"), "C1", "C2")
```

# 6. Triangle Centers

## 6.1. Basic Centers

### 6.1.1. Centroid — `ctz-def-centroid`

The intersection of medians (center of mass):

**Code**

```
#ctz-canvas(length: 0.8cm, {
  import cetz.draw: *
  ctz-init()

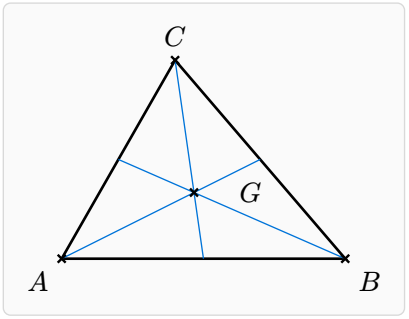
  ctz-def-points(A: (0, 0), B: (5, 0), C: (2, 3.5))
  ctz-def-centroid("G", "A", "B", "C")

  // Draw medians
  ctz-def-midpoint("Ma", "B", "C")
  ctz-def-midpoint("Mb", "A", "C")
  ctz-def-midpoint("Mc", "A", "B")

  ctz-draw(line: ("A", "B", "C", "A"), stroke: black)
  ctz-draw(segment: ("A", "Ma"), stroke: blue +
0.5pt)
  ctz-draw(segment: ("B", "Mb"), stroke: blue +
0.5pt)
  ctz-draw(segment: ("C", "Mc"), stroke: blue +
0.5pt)

  ctz-draw(points: ("A", "B", "C", "G"), labels: (
    A: "below left",
    B: "below right",
    C: "above",
    G: "above right"
  ))
})
```

**Figure**




### 6.1.2. Circumcenter — `ctz-def-circumcenter`

Center of the circumscribed circle:

**Code**

```
#ctz-canvas(length: 0.75cm, clip-canvas: (-0.5, -0.5,
5.5, 4), {
  import cetz.draw: *
  ctz-init()

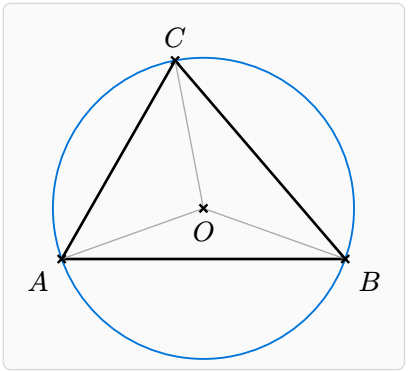
  ctz-def-points(A: (0, 0), B: (5, 0), C: (2, 3.5))
  ctz-def-circumcenter("O", "A", "B", "C")

  ctz-draw(line: ("A", "B", "C", "A"), stroke: black)
  ctz-draw(circle-through: ("O", "A"), stroke: blue +
0.7pt)

  ctz-draw(segment: ("O", "A"), stroke: gray + 0.5pt)
  ctz-draw(segment: ("O", "B"), stroke: gray + 0.5pt)
  ctz-draw(segment: ("O", "C"), stroke: gray + 0.5pt)

  ctz-draw(points: ("A", "B", "C", "O"), labels: (
    A: "below left",
    B: "below right",
    C: "above",
    O: "below"
  ))
})
```

**Figure**



### 6.1.3. Incenter — `ctz-def-incenter`

Center of the inscribed circle:

**Code**

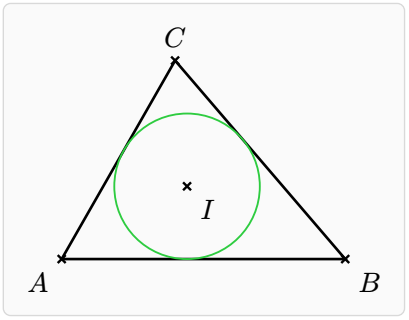
```
#ctz-canvas(length: 0.75cm, clip-canvas: (-0.5, -0.5,
5.5, 4), {
  import cetz.draw: *
  ctz-init()

  ctz-def-points(A: (0, 0), B: (5, 0), C: (2, 3.5))
  ctz-def-incenter("I", "A", "B", "C")

  ctz-draw(line: ("A", "B", "C", "A"), stroke: black)
  ctz-draw(incircle: ("A", "B", "C"), stroke: green +
0.7pt)

  ctz-draw(points: ("A", "B", "C", "I"), labels: (
    A: "below left",
    B: "below right",
    C: "above",
    I: "below right"
  ))
})
```

**Figure**




### 6.1.4. Orthocenter — `ctz-def-orthocenter`

Intersection of altitudes:

**Code**

```
#ctz-canvas(length: 0.75cm, {
  import cetz.draw: *
  ctz-init()

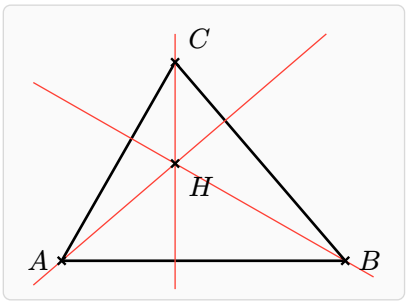
  ctz-def-points(A: (0, 0), B: (5, 0), C: (2, 3.5))
  ctz-def-orthocenter("H", "A", "B", "C")

  // Altitudes
  ctz-def-perp("Ha1", "Ha2", ("B", "C"), "A")
  ctz-def-perp("Hb1", "Hb2", ("A", "C"), "B")
  ctz-def-perp("Hc1", "Hc2", ("A", "B"), "C")

  ctz-draw(line: ("A", "B", "C", "A"), stroke: black)
  ctz-draw-line-add("A", "Ha1", add: (2, 2), stroke:
red + 0.5pt)
  ctz-draw-line-add("B", "Hb1", add: (2, 2), stroke:
red + 0.5pt)
  ctz-draw-line-add("C", "Hc1", add: (2, 2), stroke:
red + 0.5pt)

  ctz-draw(points: ("A", "B", "C", "H"), labels: (
    A: "left",
    B: "right",
    C: "above right",
    H: "below right"
  ))
})
```

**Figure**




## 6.2. The Euler Line

In any non-equilateral triangle, the orthocenter $H$, centroid $G$, and circumcenter $O$ are collinear. This line is called the **Euler line**, and remarkably, $G$ divides $HO$ in the ratio $2:1$.

**Code**

```
#ctz-canvas(length: 0.75cm, clip-canvas: (-0.5, -0.5,
5.5, 4), {
  import cetz.draw: *
  ctz-init()

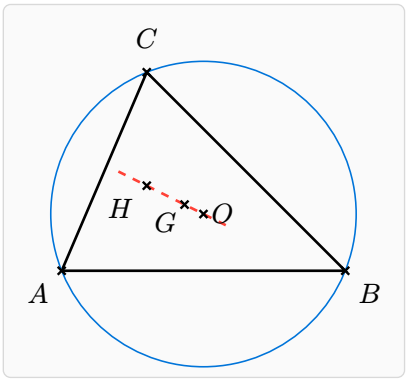
  ctz-def-points(A: (0, 0), B: (5, 0), C: (1.5, 3.5))

  ctz-def-orthocenter("H", "A", "B", "C")
  ctz-def-centroid("G", "A", "B", "C")
  ctz-def-circumcenter("O", "A", "B", "C")

  ctz-draw(line: ("A", "B", "C", "A"), stroke: black)
  ctz-draw-line-add("H", "O", add: 0.5, stroke:
(paint: red, dash: "dashed"))
  ctz-draw(circle-through: ("O", "A"), stroke: blue +
0.6pt)

  ctz-draw(points: ("A", "B", "C", "H", "G", "O"),
labels: (
    A: "below left",
    B: "below right",
    C: "above",
    H: "left",
    G: "below",
    O: "right"
  ))
})
```

**Figure**




## 6.3. Right Triangles via Thales' Theorem

Thales' theorem states that any triangle inscribed in a semicircle with the diameter as its base has a right angle at the opposite vertex. The `ctz-def-thales-triangle()` function creates such triangles.

**Code**

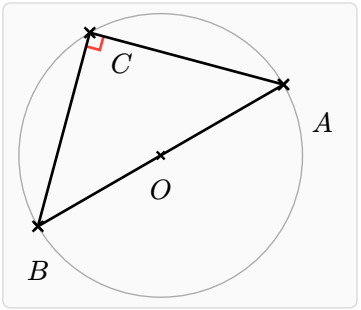
```
#ctz-canvas(length: 0.75cm, {
  import cetz.draw: *
  ctz-init()

  ctz-def-points(O: (0, 0))
  ctz-def-thales-triangle("A", "B", "C", "O", 2.5,
    base-angle: 30, orientation: "left")

  ctz-draw(circle-r: ("O", 2.5), stroke: gray +
0.5pt)
  ctz-draw-path("A--B--C--A", stroke: black + 1pt)
  ctz-draw-mark-right-angle("A", "C", "B", color:
red)

  ctz-draw(points: ("A", "B", "C", "O"), labels: (
    A: "right",
    B: "left",
    C: "above",
    O: "below"
  ))
})
```

**Figure**




Parameters:

- `name-a, name-b`: Diameter endpoints (base of triangle)
- `name-c`: Vertex with the right angle
- `center`: Circle center
- `radius`: Circle radius
- `base-angle`: Rotation angle for the diameter (default: 0)
- `orientation`: "left" or "right" - position of right angle vertex

## 6.4. Advanced Centers

`ctz-euclide` supports 10+ specialized triangle centers:

- `ctz-def-lemoine` — Symmedian point (Lemoine point)
- `ctz-def-nagel` — Nagel point
- `ctz-def-gergonne` — Gergonne point
- `ctz-def-spieker` — Spieker center (incenter of medial triangle)
- `ctz-def-euler` — Nine-point circle center
- `ctz-def-feuerbach` — Feuerbach point
- `ctz-def-mittenpunkt` — Mittenpunkt
- `ctz-def-excenter` — Excenter (specify vertex: `"a"`, `"b"`, or `"c"`)

Example with Euler (nine-point) circle:

**Code**

```
#ctz-canvas(length: 0.7cm, {
  import cetz.draw: *
  ctz-init()

  ctz-def-points(A: (0, 0), B: (5, 0), C: (1.5, 3.5))
  ctz-def-euler("N", "A", "B", "C")

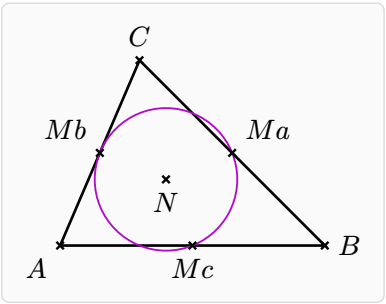
  ctz-draw(line: ("A", "B", "C", "A"), stroke: black)

  // Nine-point circle passes through midpoints
  ctz-def-midpoint("Ma", "B", "C")
  ctz-def-midpoint("Mb", "A", "C")
  ctz-def-midpoint("Mc", "A", "B")

  ctz-draw(circle-through: ("N", "Ma"), stroke:
purple + 0.7pt)

  ctz-draw(points: ("A", "B", "C", "N", "Ma", "Mb",
"Mc"), labels: (
    N: "below"
  ))
})
```

**Figure**



# 7. Transformations

## 7.1. Rotation — `rotate`

Rotate a point around a center:

**Code**

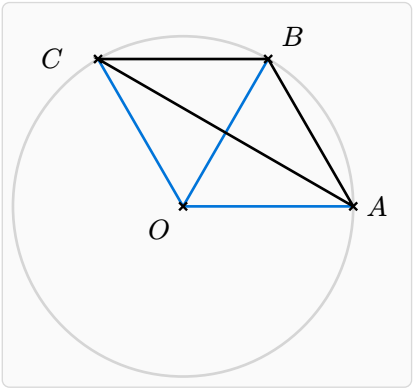
```
#ctz-canvas(length: 0.75cm, {
  import cetz.draw: *
  ctz-init()

  ctz-def-points(O: (2, 2), A: (5, 2))
  ctz-def-rotation("B", "A", "O", 60)
  ctz-def-rotation("C", "A", "O", 120)

  ctz-draw(circle-r: ("O", 3), stroke:
gray.lighten(50%))
  ctz-draw(segment: ("O", "A"), stroke: blue)
  ctz-draw(segment: ("O", "B"), stroke: blue)
  ctz-draw(segment: ("O", "C"), stroke: blue)
  ctz-draw(line: ("A", "B", "C", "A"), stroke: black)

  ctz-draw(points: ("O", "A", "B", "C"), labels: (
    O: "below left",
    A: "right",
    B: "above right",
    C: "left"
  ))
})
```

**Figure**




## 7.2. Reflection — `ctz-def-reflect`

Reflect a point across a line:

**Code**

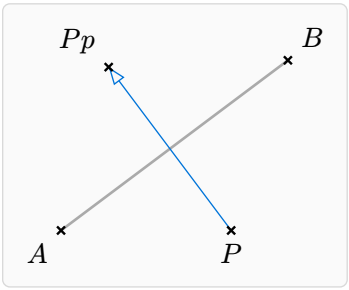
```
#ctz-canvas(length: 0.75cm, {
  import cetz.draw: *
  ctz-init()

  ctz-def-points(A: (0, 0), B: (4, 3), P: (3, 0))
  ctz-def-reflect("Pp", "P", "A", "B")

  ctz-draw(segment: ("A", "B"), stroke: gray)
  ctz-draw(path: "P--Pp", stroke: blue + 0.5pt, mark:
(end: ">"), points: false, labels: false)

  ctz-draw(points: ("A", "B", "P", "Pp"), labels: (
    P: "below",
    Pp: "above left"
  ))
})
```

**Figure**




## 7.3. Homothety (Scaling) — `scale`

Scale a point from a center:

**Code**

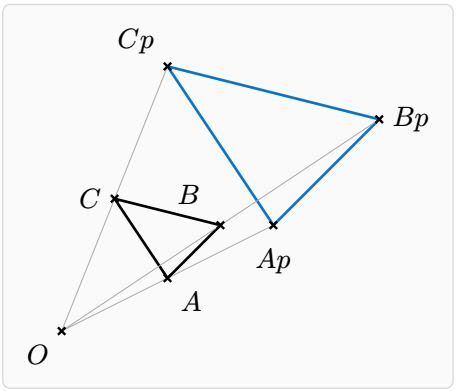
```
#ctz-canvas(length: 0.7cm, {
  import cetz.draw: *
  ctz-init()

  ctz-def-points(O: (0, 0), A: (2, 1), B: (3, 2), C:
(1, 2.5))
  ctz-def-homothety("Ap", "A", "O", 2)
  ctz-def-homothety("Bp", "B", "O", 2)
  ctz-def-homothety("Cp", "C", "O", 2)

  ctz-draw(line: ("A", "B", "C", "A"), stroke: black)
  ctz-draw(line: ("Ap", "Bp", "Cp", "Ap"), stroke:
blue)
  ctz-draw(segment: ("O", "Ap"), stroke: gray +
0.3pt)
  ctz-draw(segment: ("O", "Bp"), stroke: gray +
0.3pt)
  ctz-draw(segment: ("O", "Cp"), stroke: gray +
0.3pt)

  ctz-draw(points: ("O", "A", "B", "C", "Ap", "Bp",
"Cp"), labels: (
    O: "below left"
  ))
})
```

**Figure**




## 7.4. Projection — `ctz-def-project`

Project a point onto a line:

**Code**

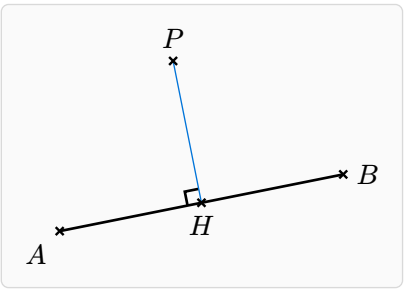
```
#ctz-canvas(length: 0.75cm, {
  import cetz.draw: *
  ctz-init()

  ctz-def-points(A: (0, 0), B: (5, 1), P: (2, 3))
  ctz-def-project("H", "P", "A", "B")

  ctz-draw(segment: ("A", "B"), stroke: black)
  ctz-draw(segment: ("P", "H"), stroke: blue + 0.5pt)
  ctz-draw-mark-right-angle("A", "H", "P", size:
0.25)

  ctz-draw(points: ("A", "B", "P", "H"), labels: (
    P: "above",
    H: "below"
  ))
})
```

**Figure**




## 7.5. Inversion — `ctz-def-inversion`

Invert points, lines, or circles through a circle:

**Code**

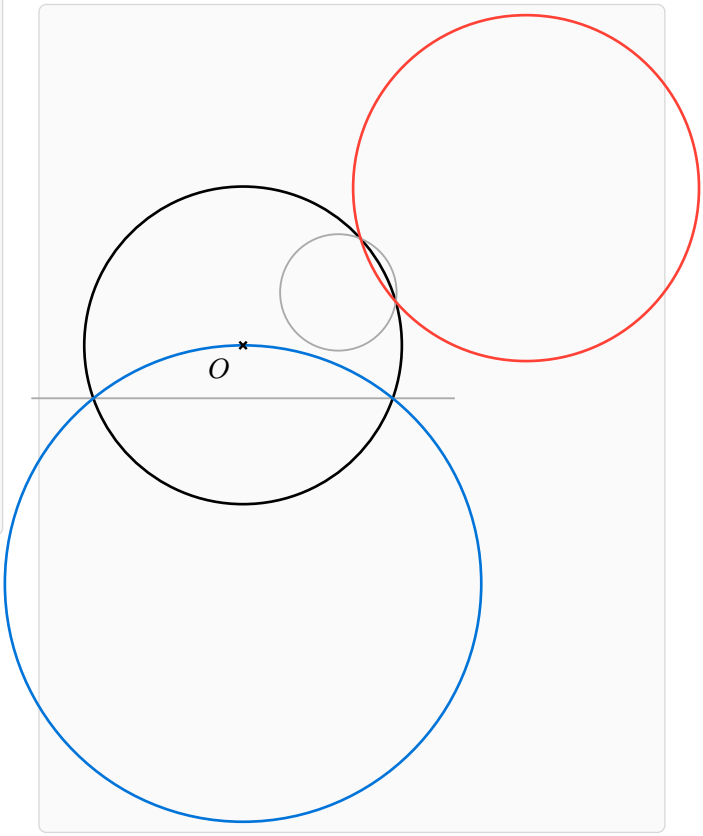
```
#ctz-canvas(length: 0.7cm, clip-canvas: (-5, -4.5, 5,
4), {
  import cetz.draw: *
  ctz-init()

  ctz-def-points(O: (0, 0), A: (-4, -1), B: (4, -1),
C: (1.8, 1))
  ctz-def-line("L", "A", "B")
  ctz-def-circle("C1", "C", radius: 1.1)

  ctz-def-inversion("Li", "L", "O", 3)
  ctz-def-inversion("C1i", "C1", "O", 3)

  ctz-draw(circle-r: ("O", 3), stroke: black + 1pt)
  ctz-draw("L", stroke: gray + 0.7pt)
  ctz-draw("C1", stroke: gray + 0.7pt)
  ctz-draw("Li", stroke: blue + 1pt)
  ctz-draw("C1i", stroke: red + 1pt)
  ctz-draw(points: ("O"), labels: (O: "below left"))
})
```

**Figure**




## 7.6. Object Duplication — `ctz-duplicate`

Duplicate any geometric object. For polygons, explicit point names must be provided.

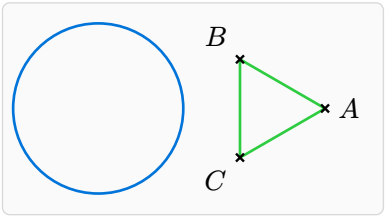
**Code**

```
#ctz-canvas(length: 0.75cm, {
  import cetz.draw: *
  ctz-init()

  // Duplicate a circle
  ctz-def-circle("c1", (0, 0), radius: 1.5)
  ctz-duplicate("c2", "c1")
  ctz-draw("c1", stroke: blue)

  // Duplicate a polygon
  ctz-def-regular-polygon("tri", ("A","B","C"),
    (3, 0), (4, 0), n: 3)
  ctz-duplicate("tri2", "tri",
    points: ("A2", "B2", "C2"))
  ctz-draw("tri", stroke: green)
  ctz-draw(points: ("A", "B", "C"), labels: true)
})
```

**Figure**




For points, lines, and circles, duplication is straightforward. For polygons, you must provide explicit point names so the vertices can be referenced independently.

## 7.7. Polymorphic Rotation

The `ctz-def-rotation()` function works on all object types: points, lines, circles, and polygons.

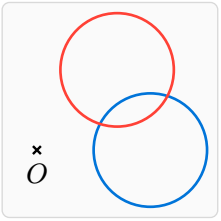
**Code**

```
#ctz-canvas(length: 0.75cm, {
  import cetz.draw: *
  ctz-init()

  // Rotate a circle
  ctz-def-points(O: (0, 0))
  ctz-def-circle("c1", (2, 0), radius: 1)
  ctz-def-rotation("c2", "c1", "O", 45)

  ctz-draw("c1", stroke: blue)
  ctz-draw("c2", stroke: red)
  ctz-draw(points: ("O"), labels: (
    O: "below"
  ))
})
```

**Figure**




For lines, both endpoints are rotated. For circles, the center is rotated while radius remains constant. For polygons, all constituent points are rotated in place.

# 8. Drawing & Styling

## 8.1. Global Style Configuration

The package provides configurable style variables that control the default appearance of points, crosses, and marks. These are defined in `draw.typ` and can be overridden.

### 8.1.1. Style Variables

| Variable | Default | Description |
|---|---|---|
| `default-point-stroke-width` | `0.9pt` | Stroke width for point markers (crosses, plus signs) |
| `default-mark-stroke-width` | `1pt` | Stroke width for tick marks on segments and arcs |
| `default-point-size` | `0.07` | Point size in canvas units (for `ctz-draw` points) |
| `default-point-size-pt` | `2pt` | Point size in pt (for path markers like crosses) |
| `default-point-size-small-pt` | `1.5pt` | Smaller point size in pt (for dots, circles, squares) |
| `default-point-color` | `black` | Default color for points |
| `default-line-color` | `black` | Default color for lines and segments |
| `default-mark-color` | `black` | Default color for construction marks |
| `default-point-shape` | `"cross"` | Default point shape (cross, dot, circle, plus, square, diamond, triangle) |

### 8.1.2. Customizing Styles

To customize the default appearance, modify the variables in `src/draw.typ` at the top of the file:

```
// In src/draw.typ - modify these values:
#let default-point-stroke-width = 1.2pt   // thicker crosses
#let default-mark-stroke-width = 1.5pt    // thicker tick marks
#let default-point-size = 0.1             // larger points
#let default-point-size-pt = 3pt          // larger path markers
#let default-point-color = blue           // blue points
#let default-line-color = gray            // gray lines
#let default-mark-color = red             // red tick marks
#let default-point-shape = "dot"          // use dots instead of crosses
```

All drawing functions throughout the package reference these variables, so changing them once affects all drawings consistently.

The public API also exports these as `ctz-default-point-stroke-width`, `ctz-default-point-color`, etc., for inspection.

## 8.2. Automatic Label Placement (Collision Avoidance)

Labels are placed **as close as possible to the requested anchor** while avoiding everything that has already been drawn: segments, circles, point markers, and other labels. This is **on by default**.

The engine works per label, deterministically:

1. Measure the label and build its bounding box (including the standard 5pt gap around the text). In tight clusters — another point closer than the label could ever be — the gap is reduced to 60% so the label hugs its point instead of floating; points in open space keep the airy gap.
2. Try candidate positions in order of visual "closeness" to your request: the requested anchor first (pushed slightly farther out if needed), then a 45° lean, then the **opposite** side — flipping above ↔ below keeps the label centered on the same axis, so a crowded row naturally alternates its labels above and below the line. Perpendicular and diagonal flips come last. When the anchor grid finds nothing nearby, **free-floating** candidates at 16 fine directions around the point take over — they can nestle a label into a tilted

wedge (e.g. between two medians crossing at a centroid). **Every** full-size position is tried before the font is shrunk (85%, then 70%) — shrinking is a last resort.
3. A candidate is accepted when its **text** is clear of everything: thin geometry may run through the label's padding (a close, slightly grazed label reads better than one exiled to the first perfectly clear spot). If nothing is acceptable, the least-bad candidate is used (never an unbounded search).
4. If any label of a figure must shrink, **all** labels of that figure are re-placed at the same (smallest needed) scale — mixed sizes look accidental. The scale also applies to labels drawn later in the canvas.
5. **Group tidiness:** if even one label had to travel far from its point (staggered rows, big anchor flips) or still collides, the figure is overcrowded for its label size. The engine then retries the whole figure at a uniform smaller scale and keeps the largest scale where the layout is **calm** — every label at its requested anchor, nothing colliding. One quiet row of slightly smaller labels reads better than full-size chaos; if no scale is calm, the full-size layout is kept.

When the engine is active, labels without an explicit position are fully automatic (`labels: true` picks each label's side from the local geometry). `"auto"` also works per label, so you can mix automatic, suggested and pinned labels in one call:

```
ctz-draw(points: ("A", "B", "C"), labels: (
  A: "auto",                 // engine picks the side from the geometry
  B: "left",                 // suggestion — may be adjusted
  C: "below+0.4 right+0.3",  // pinned exactly (see below)
))
```

Labels never overlap the **stroke** of a circle but may sit inside it, and a label placed by the engine is registered so later labels avoid it too.

**Association guarantee:** an automatically placed label is always strictly closer to its own point than to any other drawn point (`assoc-factor`, default 1.0; raise it to demand a clear margin) — candidates that would visually attach to a neighbor are rejected outright, in the fallback too.

**Important:** the engine only sees geometry drawn **before** the label — draw your labels last (all examples in this manual already do).

#### 8.2.1. Turning it off and on

```
ctz-label-avoid(false)   // global off: labels go exactly where asked
ctz-label-avoid(true)    // back on with defaults
```

Per label, `avoid: false` pins the label exactly where requested:

```
ctz-draw(points: ("A", "B", "M"), labels: (
  A: "below left",
  M: (pos: "below", avoid: false),  // exact position, no dodging
  B: "below right",
))
```

`ctz-label-avoid(...)` can be called mid-canvas; it affects all labels drawn after it.

#### 8.2.2. Precise manual placement

A plain position name (`"above"`) is a **suggestion** — the engine may move the label to keep it readable. To place a label **exactly**, add numeric offsets (canvas units) to direction tokens; such a label is pinned as written and the engine ignores it:

```
ctz-draw(points: ("A", "B", "M"), labels: (
  A: "above+1.2 left+0.5",   // 1.2 up, 0.5 left of A — exactly there
  B: "top+0.8",              // top/bottom/up/down are aliases
```

```
  M: "below+0.3 right+0.2",
))
```

The direction tokens also set the text anchor (`"above+1.2 left+0.5"` anchors like `"above left"`, so the text extends away from the point), and the standard 5pt text gap still applies — `"above+0"` is exactly a pinned `"above"`. A negative amount flips the direction (`"left-0.5"` moves right). The equivalent dictionary form is `(pos: "above left", offset: (-0.5, 1.2), avoid: false)`.

### 8.2.3. Configuration

Pass a dictionary to `ctz-label-avoid` (or set `ctz-style(label: (avoid: (...)))`) to tune the engine. All distance options accept Typst lengths (canvas-scale independent) or plain numbers (canvas units).

| Option | Default | Meaning |
|---|---|---|
| `ring-step` | `4pt` | Extra distance per push-out retry |
| `rings` | `5` | Number of distances tried per anchor |
| `allow-shrink` | `true` | `false`: never reduce the font size — overcrowded figures stagger their labels at full size instead |
| `shrink` | `(0.85, 0.7)` | Font scales tried when full size does not fit calmly |
| `assoc-factor` | `1.0` | A label must be this factor closer to its own point than to any other (1.0 = strictly closer) |
| `margin` | `1.5pt` | Clearance kept around points and circle strokes |

```
ctz-label-avoid((allow-shrink: false))   // full-size labels, whatever happens
ctz-label-avoid((rings: 8))              // search farther out
```

Angle labels (`ctz-draw-angle`) participate too: they slide outward along the angle bisector until clear, and register their footprint so point labels flow around them.

## 8.3. Main Levée (Hand-Drawn Style)

The package supports a "main levée" (hand-drawn/sketchy) style that adds slight perturbations to lines and shapes, giving them a natural, hand-drawn appearance.

### 8.3.1. Main Levée Variables

| Variable | Default | Description |
|---|---|---|
| `default-main-levee` | `false` | Enable hand-drawn style globally |
| `default-main-levee-roughness` | `1.0` | Roughness amount (0 = smooth, 1-2 = typical) |
| `default-main-levee-seed` | `42` | Seed for reproducible randomness |

### 8.3.2. Using Sketchy Mode with `ctz-draw`

The simplest way to use hand-drawn style is to add `sketchy: true` to any `ctz-draw()` call. This works with segments, circles, polylines, ellipses, arcs, and more.

#### 8.3.2.1. Sketchy Segments

**Code**

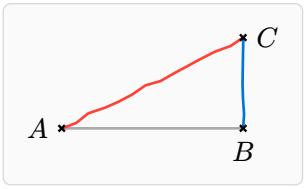
```
#ctz-canvas(length: 0.6cm, {
  import cetz.draw: *
  ctz-init()
  ctz-def-points("A", (0, 0), "B", (4, 0), "C", (4,
2))

  // Normal segment for comparison
  ctz-draw(segment: ("A", "B"), stroke: gray)

  // Sketchy segments with different roughness
  ctz-draw(segment: ("B", "C"), stroke: blue,
sketchy: true)
  ctz-draw(segment: ("C", "A"), stroke: red, sketchy:
true, roughness: 1.5)

  ctz-draw(points: ("A", "B", "C"), labels: (A:
"left", B: "below", C: "right"))
})
```

**Figure**




#### 8.3.2.2. Sketchy Circles

**Code**

```
#ctz-canvas(length: 0.5cm, {
  import cetz.draw: *
  ctz-init()
  ctz-def-points("O", (0, 0), "P", (1.5, 0), "O2",
(5, 0), "P2", (6.5, 0))

  // Normal circle
  ctz-draw(circle-through: ("O", "P"), stroke: gray)

  // Sketchy circle
  ctz-draw(circle-through: ("O2", "P2"), stroke:
blue, sketchy: true)

  ctz-draw(points: ("O", "O2"))
})
```

**Figure**



#### 8.3.2.3. Sketchy Polylines (Triangles)

**Code**

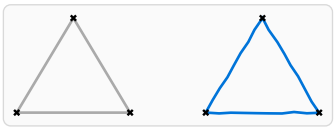
```
#ctz-canvas(length: 0.5cm, {
  import cetz.draw: *
  ctz-init()
  ctz-def-points("A", (0, 0), "B", (3, 0), "C", (1.5,
2.5))
  ctz-def-points("A2", (5, 0), "B2", (8, 0), "C2",
(6.5, 2.5))

  // Normal triangle
  ctz-draw(line: ("A", "B", "C", "A"), stroke: gray)

  // Sketchy triangle
  ctz-draw(line: ("A2", "B2", "C2", "A2"), stroke:
blue, sketchy: true)

  ctz-draw(points: ("A", "B", "C", "A2", "B2", "C2"))
})
```

**Figure**



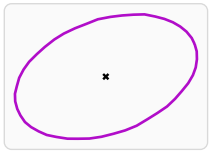
#### 8.3.2.4. Sketchy Ellipse

**Code**

```
#ctz-canvas(length: 0.5cm, {
  import cetz.draw: *
  ctz-init()
  ctz-def-points("O", (0, 0))

  ctz-draw(ellipse: ("O", 2.5, 1.5, 20deg), stroke:
purple, sketchy: true)
  ctz-draw(points: ("O",))
})
```

**Figure**



#### 8.3.2.5. Sketchy Circumcircle and Incircle

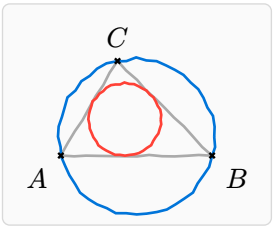
**Code**

```
#ctz-canvas(length: 0.5cm, {
  import cetz.draw: *
  ctz-init()
  ctz-def-points("A", (0, 0), "B", (4, 0), "C", (1.5,
2.5))

  ctz-draw(line: ("A", "B", "C", "A"), stroke: gray,
sketchy: true)
  ctz-draw(circumcircle: ("A", "B", "C"), stroke:
blue, sketchy: true)
  ctz-draw(incircle: ("A", "B", "C"), stroke: red,
sketchy: true)

  ctz-draw(points: ("A", "B", "C"), labels: (
    A: "below left", B: "below right", C: "above"))
})
```

**Figure**



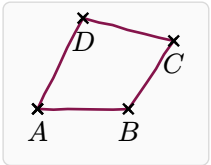

#### 8.3.2.6. Sketchy Path

**Code**

```
#ctz-canvas(length: 0.6cm, {
  import cetz.draw: *
  ctz-init()
  ctz-def-points("A", (0, 0), "B", (2, 0), "C", (3,
1.5), "D", (1, 2))

  ctz-draw(path: "A--B--C--D--A", stroke: maroon,
sketchy: true, roughness: 0.8)
  ctz-draw(points: ("A", "B", "C", "D"))
})
```

**Figure**




### 8.3.3. Sketchy Parameters

The following parameters control the hand-drawn appearance:

- `sketchy: true` — Enable hand-drawn style
- `roughness: 1.0` — Controls wobbliness (0 = smooth, 1-2 = typical hand-drawn)
- `seed: 42` — Seed for reproducible randomness (same seed = same wobbles)

### 8.3.4. Roughness Comparison

The `roughness` parameter controls how “wobbly” the lines are:

**Code**

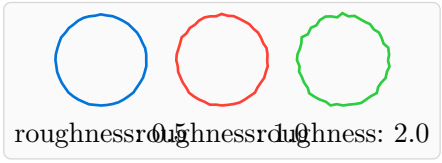
```
#ctz-canvas(length: 0.4cm, {
  import cetz.draw: *
  ctz-init()
  ctz-def-points("O1", (-4, 0), "P1", (-2.5, 0))
  ctz-def-points("O2", (0, 0), "P2", (1.5, 0))
  ctz-def-points("O3", (4, 0), "P3", (5.5, 0))

  // roughness = 0.5 (subtle)
  ctz-draw(circle-through: ("O1", "P1"), stroke:
blue, sketchy: true, roughness: 0.5)
  content((-4, -2.5), [roughness: 0.5])

  // roughness = 1.0 (default)
  ctz-draw(circle-through: ("O2", "P2"), stroke: red,
sketchy: true, roughness: 1.0)
  content((0, -2.5), [roughness: 1.0])

  // roughness = 2.0 (very sketchy)
  ctz-draw(circle-through: ("O3", "P3"), stroke:
green, sketchy: true, roughness: 2.0)
  content((4, -2.5), [roughness: 2.0])
})
```

**Figure**




### 8.3.5. Supported Constructs

The `sketchy: true` parameter works with:

| Construct | Example |
|---|---|
| `segment:` | `ctz-draw(segment: ("A", "B"), sketchy: true)` |
| `line:` | `ctz-draw(line: ("A", "B", "C"), sketchy: true)` |
| `path:` | `ctz-draw(path: "A--B--C", sketchy: true)` |
| `circle-through:` | `ctz-draw(circle-through: ("O", "P"), sketchy: true)` |
| `circle-r:` | `ctz-draw(circle-r: ("O", 2), sketchy: true)` |
| `circle-diameter:` | `ctz-draw(circle-diameter: ("A", "B"), sketchy: true)` |
| `circumcircle:` | `ctz-draw(circumcircle: ("A", "B", "C"), sketchy: true)` |
| `incircle:` | `ctz-draw(incircle: ("A", "B", "C"), sketchy: true)` |
| `ellipse:` | `ctz-draw(ellipse: ("O", 2, 1.5), sketchy: true)` |
| `arc:` | `ctz-draw(arc: (center: "O", start: "A", end: "B"), sketchy: true)` |
| `arc-r:` | `ctz-draw(arc-r: ("O", 2, 0, 90), sketchy: true)` |
| `semicircle:` | `ctz-draw(semicircle: ("A", "B"), sketchy: true)` |
| Named objects | `ctz-draw("myCircle", sketchy: true)` |

### 8.3.6. Low-Level Sketchy Functions

For more control, you can also use the dedicated sketchy functions directly:

```
ctz-sketchy-line((0, 0), (4, 0), stroke: blue, roughness: 1.0)
ctz-sketchy-circle((0, 0), 2, stroke: red, roughness: 1.2)
ctz-sketchy-polygon((0, 0), (3, 0), (1.5, 2.5), stroke: green)
ctz-sketchy-ellipse((0, 0), 2, 1.5, angle: 20deg, stroke: purple)
ctz-sketchy-rect((-2, -1), (2, 1), stroke: orange)
ctz-sketchy-arc((0, 0), 2, 0deg, 90deg, stroke: teal)
```

## 8.4. Points — `ctz-draw-points`

Draw points at named locations:

```
ctz-draw(points: ("A", "B", "C"))
```

## 8.5. Unified Drawing — `ctz-draw`

The `ctz-draw()` function provides a unified interface for drawing both **named objects** and **unnamed constructs**.

### 8.5.1. Drawing Named Objects

Use `ctz-draw()` to draw any object type without remembering type-specific commands. It automatically detects whether the object is a point, line, circle, polygon, or conic (ellipse/parabola/projectile).

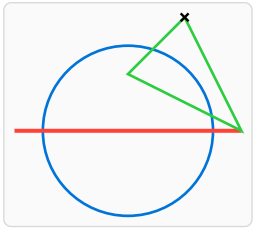
**Code**

```
#ctz-canvas(length: 0.75cm, {
  import cetz.draw: *
  ctz-init()

  ctz-def-circle("c1", (0, 0), radius: 1.5)
  ctz-def-line("l1", (-2, 0), (2, 0))
  ctz-def-polygon("tri", "A", "B", "C")
  ctz-def-points(A: (1, 2), B: (2, 0), C: (0, 1))

  ctz-draw("c1", stroke: blue, fill: none)
  ctz-draw("l1", stroke: red + 1.5pt)
  ctz-draw("tri", stroke: green)
  ctz-draw("A")
})
```

**Figure**



### 8.5.2. Drawing Unnamed Constructs

You can also use `ctz-draw()` to draw geometric objects directly without defining them first using named parameters.

**Define vs Draw**: Most `ctz-draw()` type parameters have a corresponding `ctz-def-*` function. Use `ctz-def-*` when you need to reuse the object (for intersections, transformations, etc.). Use the draw shorthand for one-off drawing. See the summary table at the end of this section.

#### 8.5.2.1. Points with Labels

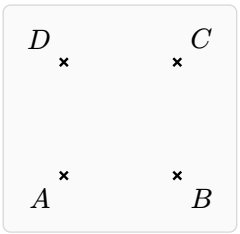
Draw multiple points at once, optionally with labels:

**Code**

```
#ctz-canvas(length: 0.75cm, {
  import cetz.draw: *
  ctz-init()
  ctz-def-points(A: (0, 0), B: (2, 0), C: (2, 2), D:
(0, 2))

  // Draw points with custom label positions
  ctz-draw(points: ("A", "B", "C", "D"), labels: (
    A: "below left",
    B: "below right",
    C: "above right",
    D: "above left"
  ))
})
```

**Figure**




Use `labels: true` for default label positioning, or omit the `labels` parameter to draw points without labels.

#### 8.5.2.2. Paths and Polylines

Draw paths using the `path:` parameter for CeTZ-style path syntax, or `line:` for polylines through points:

```
ctz-draw(path: "A--B--C--A", stroke: black)  // Close the path by repeating first point
ctz-draw(line: ("A", "B", "C", "D"), stroke: red)  // Open polyline
```

#### 8.5.2.3. Circles

Draw circles without naming them:

```
// Circle through two points (center and point on circumference)
ctz-draw(circle-through: ("O", "A"), stroke: blue)

// Circle by center and radius
ctz-draw(circle-r: ((0, 0), 1.5), stroke: green)

// Circle by diameter endpoints
ctz-draw(circle-diameter: ("A", "B"), stroke: purple)

// Circumcircle of triangle
ctz-draw(circumcircle: ("A", "B", "C"), stroke: red)

// Incircle of triangle
ctz-draw(incircle: ("A", "B", "C"), stroke: teal)
```

#### 8.5.2.4. Conics (Ellipses, Parabolas, Projectiles)

Draw conics without naming them:

```
// Ellipse by center, radii, and rotation angle
ctz-draw(ellipse: ((0, 0), 3, 2, 20deg), stroke: black)

// Parabola by focus + directrix (line or two points)
ctz-draw(parabola: ((0, 0), ((-2, -3), (-2, 3)), 4), stroke: black)

// Projectile by origin + velocity (optional gravity, vectors, etc.)
ctz-draw(projectile: (origin: (0, 0), velocity: (4, 6), y-floor: 0, vectors: true), stroke:
blue)
```

#### 8.5.2.5. Arcs and Semicircles

```
// Arc by center and two points
ctz-draw(arc: (center: "O", start: "A", end: "B"), stroke: orange)

// Arc by center, radius, and angles (in degrees)
ctz-draw(arc-r: ((0, 0), 2, 0, 90), stroke: black)

// Semicircle by diameter endpoints
ctz-draw(semicircle: ("A", "B"), stroke: blue, fill: blue.lighten(80%))
```

#### 8.5.2.6. Line Segments

```
ctz-draw(segment: ("A", "B"), stroke: maroon + 1.5pt)
```

### 8.5.3. Complete Example

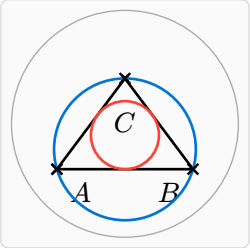
**Code**

```
#ctz-canvas(length: 0.6cm, {
  import cetz.draw: *
  ctz-init()
  ctz-def-points(A: (0, 0), B: (3, 0), C: (1.5, 2))

  // Named circle
  ctz-def-circle("myCircle", (1.5, 1), radius: 2.5)
  ctz-draw("myCircle", stroke: gray + 0.5pt)

  // Unnamed constructs
  ctz-draw(path: "A--B--C--A", stroke: black)
  ctz-draw(circumcircle: ("A", "B", "C"), stroke:
blue)
  ctz-draw(incircle: ("A", "B", "C"), stroke: red)
  ctz-draw(points: ("A", "B", "C"), labels: true)
})
```

**Figure**




## 8.6. Labels — `ctz-draw-labels`

Add labels to points with positioning:

```
ctz-draw-labels("A", "B", "C",
  A: "below left",
  B: "below right",
  C: "above")
```

Positions: `"above"`, `"below"`, `"left"`, `"right"`, `"above left"`, etc.

Custom offset:

```
ctz-draw-labels("O", O: (pos: "below", offset: (0, -0.15)))
```

More placement controls (position, offset, distance):

```
ctz-draw-labels("A", "B", "C",
  A: (pos: "above", dist: 0.25),
  B: (pos: "right", offset: (0.1, 0)),
  C: (pos: "below left", offset: (-0.05, -0.05)))
```

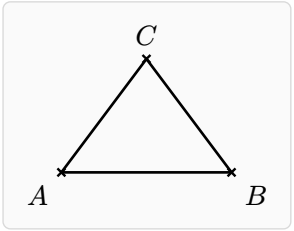
## 8.7. Labels & Points — Unified API (Recommended)

The modern approach combines point drawing and labeling in a single call:

**Code**

```
#ctz-canvas(length: 0.75cm, {
  import cetz.draw: *
  ctz-init()
  ctz-def-points(A: (0, 0), B: (3, 0), C: (1.5, 2))
  ctz-draw(line: ("A", "B", "C", "A"), stroke: black)
  ctz-draw(points: ("A", "B", "C"), labels: (
    A: "below left",
    B: "below right",
    C: "above"
  ))
})
```

**Figure**




This unified API replaces separate `ctz-draw-points()` and `ctz-draw-labels()` calls. The old API remains supported for backward compatibility.

You can also label points that were drawn earlier:

```
ctz-draw(points: ("A", "B"))  // Draw points without labels
// ... other drawing commands ...
ctz-draw(labels: (A: "below", B: "above"))  // Add labels later
```

## 8.8. Segments — `ctz-draw-segment`

Draw a segment with optional arrow or bar tips and a dimension label:

```
ctz-draw-segment("A", "B", arrows: "|-|", dim: $5$, dim-pos: "above")
```

Supported `arrows`: `--` (none), `->`, `<-`, `<->`, `|-|`, `|->`, `<-|`.

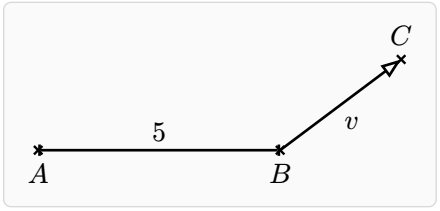
**Code**

```
#ctz-canvas(length: 0.8cm, {
  import cetz.draw: *
  ctz-init()

  ctz-def-points(A: (0, 0), B: (4, 0), C: (6, 1.5))

  ctz-draw-segment("A", "B", arrows: "|-|", dim: $5$,
dim-pos: "above")
  ctz-draw-segment("B", "C", arrows: "->", dim: $v$,
dim-pos: "below")
  ctz-draw(points: ("A", "B", "C"), labels: (A:
"below", B: "below", C: "above"))
})
```

**Figure**




Mark equal-length segments with ticks:

```
ctz-draw-mark-segment("A", "B", mark: 1)
ctz-draw-mark-segment("C", "D", mark: 2)
```

**Code**

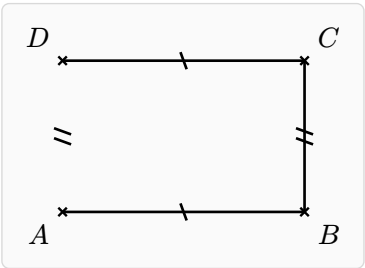
```
#ctz-canvas(length: 0.8cm, {
  import cetz.draw: *
  ctz-init()

  ctz-def-points(A: (0, 0), B: (4, 0), C: (4, 2.5),
D: (0, 2.5))
  ctz-draw(line: ("A", "B", "C", "D"), stroke: black)

  // Opposite sides equal
  ctz-draw-mark-segment("A", "B", mark: 1)
  ctz-draw-mark-segment("C", "D", mark: 1)
  ctz-draw-mark-segment("B", "C", mark: 2)
  ctz-draw-mark-segment("D", "A", mark: 2)

  ctz-draw(points: ("A", "B", "C", "D"), labels: (
    A: "below left",
    B: "below right",
    C: "above right",
    D: "above left"
  ))
})
```

**Figure**



**Code**

```
#ctz-canvas(length: 0.7cm, {
  import cetz.draw: *
  ctz-init()

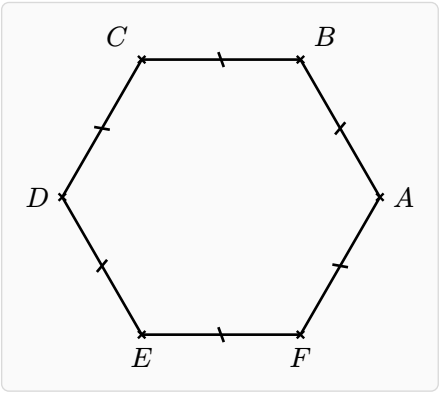
  ctz-def-points(O: (0, 0), A: (3, 0))
  ctz-def-regular-polygon("Hex", ("A", "B", "C", "D",
"E", "F"), "O", "A")

  ctz-draw-regular-polygon(("A", "B", "C", "D", "E",
"F"),
    stroke: black, mark: 1)

  // Mark all sides with the same tick
  // (use mark-opts to customize size/position)

  ctz-draw(points: ("A", "B", "C", "D", "E", "F"),
labels: (
    A: "right",
    B: "above right",
    C: "above left",
    D: "left",
    E: "below",
    F: "below"
  ))
})
```

**Figure**




## 8.9. Segment Measurements — `ctz-draw-measure-segment`

Draw an offset measurement line with dotted fences and a centered label. The line breaks around the label and uses open arrowheads by default.

```
ctz-draw-measure-segment("A", "B", label: $5$, offset: 0.3, side: "left")
```

**Code**

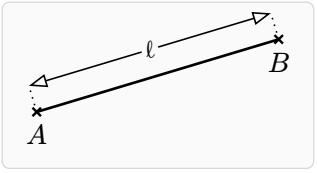
```
#ctz-canvas(length: 0.8cm, {
  import cetz.draw: *
  ctz-init()

  ctz-def-points(A: (0, 0), B: (4, 1.2))
  ctz-draw-segment("A", "B", stroke: black + 1pt)

  // Minimal measurement example
  ctz-draw-measure-segment("A", "B", label: $ell$,
offset: 0.45, side: "left",
    fence-dash: "dotted")

  ctz-draw(points: ("A", "B"), labels: (
    A: "below",
    B: "below"
  ))
})
```

**Figure**



**Code**

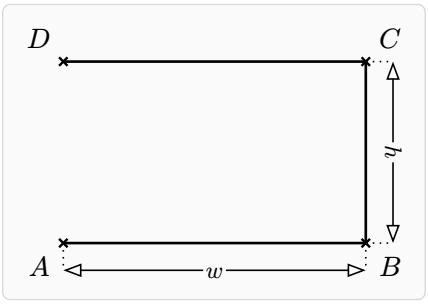
```
#ctz-canvas(length: 0.8cm, {
  import cetz.draw: *
  ctz-init()

  ctz-def-points(A: (0, 0), B: (5, 0), C: (5, 3), D:
(0, 3))
  ctz-draw(line: ("A", "B", "C", "D"), stroke: black
+ 1pt)

  // Rectangle measurements (width and height)
  ctz-draw-measure-segment("A", "B", label: $w$,
offset: 0.45, side: "below")
  ctz-draw-measure-segment("C", "B", label: $h$,
offset: -0.45, side: "right")

  ctz-draw(points: ("A", "B", "C", "D"), labels: (
    A: "below left",
    B: "below right",
    C: "above right",
    D: "above left"
  ))
})
```

**Figure**




## 8.10. Paths — `ctz-draw-path`

Draw polylines with per-segment tips using a TikZ-like string:

```
ctz-draw-path("A--B->C|-|D", stroke: black)
```

Supported connectors: `--`, `->`, `<-`, `<->`, `|-|`, `|->`, `<-|`.

By default, `ctz-draw-path` draws points as crosses for normal segments and hides points that touch a bar connector (`|-|`, `|->`, `<-|`). Labels default to `below`. You can override per-point placements or point styles in the path with `{...}`, or via `label-overrides`.

Default behavior, label are placed below

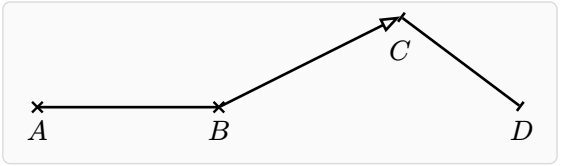
**Code**

```
#ctz-canvas(length: 0.8cm, {
  import cetz.draw: *
  ctz-init()

  ctz-def-points(
    A: (0, 0), B: (3, 0), C: (6, 1.5), D: (8, 0),
  )

  // Default labels below, default point styles
  ctz-draw-path("A--B->C|-|D", stroke: black)
})
```

**Figure**



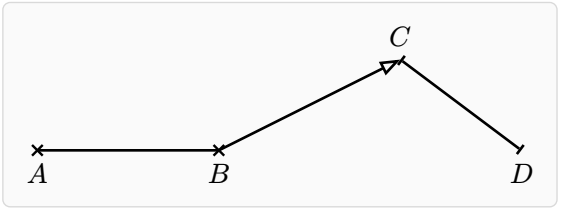
**Code**

```
#ctz-canvas(length: 0.8cm, {
  import cetz.draw: *
  ctz-init()

  ctz-def-points(
    A: (0, 0), B: (3, 0), C: (6, 1.5), D: (8, 0),
  )

  ctz-draw-path("A{below}--B{below}->C{above}|-|
D{below}", stroke: black)
})
```

**Figure**




Override placements using `label-overrides`:

```
ctz-draw-path("A--B->C|-|D",
  label-overrides: (A: "left", C: "above right"))
```

Customize point appearance or disable points/labels:

```
ctz-draw-path("A--B->C|-|D",
  point-style: "circle",
  point-color: red,
  label-pos: "above")
```

```
ctz-draw-path("A--B->C|-|D",
  points: false,
  labels: false)
```

Per-point overrides inside the path:

```
ctz-draw-path("A{below, style: circle}--B{below}->C{above, style: none}|-|D{below}",
  stroke: black)
```

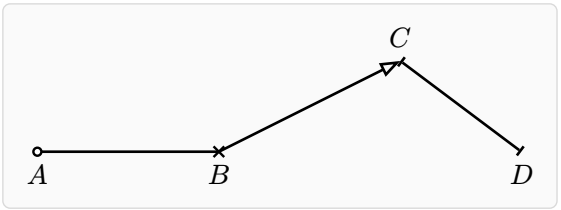
**Code**

```
#ctz-canvas(length: 0.8cm, {
  import cetz.draw: *
  ctz-init()

  ctz-def-points(
    A: (0, 0), B: (3, 0), C: (6, 1.5), D: (8, 0),
  )

  ctz-draw-path("A{below, style: circle}--B{below}-
>C{above}|-|D{below}",
    stroke: black)
})
```

**Figure**




## 8.11. Global Styling — `ctz-style`

Set default styles for points and labels:

**Code**

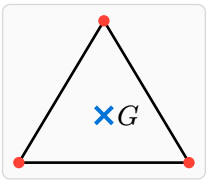
```
#ctz-canvas(length: 0.8cm, {
  import cetz.draw: *
  ctz-init()
  ctz-style(point: (shape: "dot", size: 0.1, fill:
red))

  ctz-def-points(A: (0, 0), B: (3, 0), C: (1.5, 2.5))
  ctz-draw(line: ("A", "B", "C", "A"), stroke: black)
  ctz-draw(points: ("A", "B", "C"))

  ctz-style(point: (shape: "cross", size: 0.15,
stroke: blue + 1.5pt))
  ctz-def-centroid("G", "A", "B", "C")
  ctz-draw(points: ("G"))

  ctz-draw(points: ("G"), labels: (
    A: "below left",
    B: "below right",
    C: "above",
    G: "right"
  ))
})
```

**Figure**




Point shapes: `"dot"`, `"cross"`, `"circle"`, `"square"`

## 8.12. Angle Marking — `ctz-draw-angle`

Mark and label angles:

**Code**

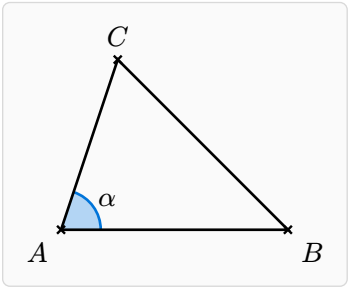
```
#ctz-canvas(length: 0.8cm, {
  import cetz.draw: *
  ctz-init()

  ctz-def-points(A: (0, 0), B: (4, 0), C: (1, 3))
  ctz-draw(line: ("A", "B", "C", "A"), stroke: black)

  ctz-draw-angle("A", "B", "C",
    label: $alpha$,
    radius: 0.7,
    fill: blue.lighten(70%),
    stroke: blue)

  ctz-draw(points: ("A", "B", "C"), labels: (
    A: "below left",
    B: "below right",
    C: "above"
  ))
})
```

**Figure**




## 8.13. Right Angle Mark — `ctz-draw-mark-right-angle`

Mark a right angle with a small square:

**Code**

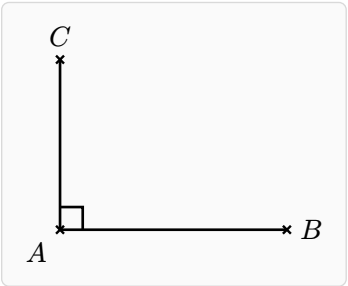
```
#ctz-canvas(length: 0.8cm, {
  import cetz.draw: *
  ctz-init()

  ctz-def-points(A: (0, 0), B: (4, 0), C: (0, 3))
  ctz-draw(segment: ("A", "B"), stroke: black)
  ctz-draw(segment: ("A", "C"), stroke: black)

  ctz-draw-mark-right-angle("B", "A", "C", size: 0.4)

  ctz-draw(points: ("A", "B", "C"), labels: (
    A: "below left",
    B: "right",
    C: "above"
  ))
})
```

**Figure**



## 8.14. Summary: Draw and Define Correspondence

The table below shows which `ctz-draw()` type parameters have corresponding `ctz-def-*` functions. Use `ctz-def-*` when you need to reference the object later (for intersections, transformations, etc.).

### 8.14.1. Geometric Objects (Have Define Functions)

| Draw Shorthand | Define Function | Object Type |
|---|---|---|
| `points: (...)` | `ctz-def-points(...)` | Points |
| `line: (...)` | `ctz-def-line(name, a, b)` | Line/Polyline |
| `circle-r: (center, r)` | `ctz-def-circle(name, center, radius: r)` | Circle |
| `circle-through: (center, pt)` | `ctz-def-circle(name, center, through: pt)` | Circle |
| `circle-diameter: (a, b)` | `ctz-def-circle-diameter(name, a, b)` | Circle |
| `circumcircle: (a, b, c)` | `ctz-def-circumcircle(name, a, b, c)` | Circle |
| `incircle: (a, b, c)` | `ctz-def-incircle(name, a, b, c)` | Circle |
| — | `ctz-def-excircle(name, a, b, c, vertex: v)` | Circle |
| — | `ctz-def-euler-circle(name, a, b, c)` | Circle |
| — | `ctz-def-spieker-circle(name, a, b, c)` | Circle |
| — | `ctz-def-apollonius-circle(name, a, b, k)` | Circle |
| `ellipse: (center, rx, ry, angle)` | `ctz-def-ellipse(name, center, rx, ry, angle: 0deg)` | Conic |
| — | `ctz-def-ellipse-foci(n1, n2, center, rx, ry, angle)` | Points (foci) |
| — | `ctz-def-ellipse-tangent(name, center, rx, ry, t, ...)` | Line (tangent) |
| — | `ctz-def-ellipse-tangents-from(n1, n2, center, rx, ry, pt, ...)` | Lines (tangents) |
| `parabola: (focus, directrix, extent)` | `ctz-def-parabola(name, focus, directrix, ...)` | Conic |
| — | `ctz-def-parabola-focus(name, focus, p, angle, ...)` | Conic |
| — | `ctz-def-parabola-tangent(name, focus, directrix, t, ...)` | Line (tangent) |
| — | `ctz-def-parabola-tangents-from(n1, n2, focus, dir, pt, ...)` | Lines (tangents) |
| `projectile: (...)` | `ctz-def-projectile(name, origin, velocity, ...)` | Conic |

### 8.14.2. Drawing Primitives (No Define Functions)

These are purely visual elements that don't need to be stored for later geometric operations:

| Draw Function/Parameter | Purpose |
|---|---|
| `arc:`, `arc-r:`, `semicircle:` | Partial circles (drawing only) |
| `segment:` | Line segment (use `ctz-def-line` if needed) |
| `path:` | CeTZ-style path syntax |

| | |
|---|---|
| `labels:` | Point labels |
| `ctz-draw-angle`, `ctz-draw-fill-angle` | Angle arcs and fills |
| `ctz-draw-mark-segment`, `ctz-draw-mark-right-angle` | Geometric marks |
| `ctz-draw-grid`, `ctz-draw-axes` | Coordinate system decoration |

# 9. Circles

This section covers all circle-related functions. Every circle drawing method has a corresponding define function, allowing you to store circles as named objects for later use (intersections, transformations, etc.).

## 9.1. Define vs Draw Pattern

For each way of creating a circle, there are two approaches:

1. **Define then Draw**: Store the circle as a named object, then draw it by name. This allows reusing the circle for intersections.
2. **Draw directly**: Use `ctz-draw()` with a type parameter to draw without storing.

```
// Approach 1: Define then draw (reusable)
ctz-def-circumcircle("C", "A", "B", "C")
ctz-draw("C", stroke: blue)
ctz-def-lc(("P", "Q"), ("A", "B"), "C")  // Can intersect with line

// Approach 2: Draw directly (one-off)
ctz-draw(circumcircle: ("A", "B", "C"), stroke: blue)
```

## 9.2. Circle by Center and Radius — `ctz-def-circle`

Define a circle by its center and either a radius value or a point on the circumference.

**Code**

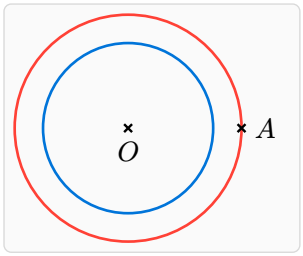
```
#ctz-canvas(length: 0.8cm, {
  import cetz.draw: *
  ctz-init()
  ctz-def-points(O: (0, 0), A: (2, 0))

  // By radius value
  ctz-def-circle("C1", "O", radius: 1.5)
  ctz-draw("C1", stroke: blue)

  // By through point
  ctz-def-circle("C2", "O", through: "A")
  ctz-draw("C2", stroke: red)

  ctz-draw(points: ("O", "A"), labels: (O: "below",
A: "right"))
})
```

**Figure**




Unnamed equivalents:

```
ctz-draw(circle-r: ((0, 0), 1.5), stroke: blue)
ctz-draw(circle-through: ("O", "A"), stroke: red)
```

## 9.3. Circle by Diameter — `ctz-def-circle-diameter`

Define a circle using two points as the diameter endpoints.

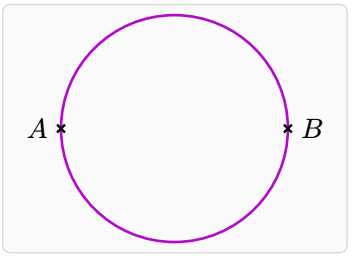
**Code**

```
#ctz-canvas(length: 0.8cm, {
  import cetz.draw: *
  ctz-init()
  ctz-def-points(A: (-2, 0), B: (2, 0))

  ctz-def-circle-diameter("C", "A", "B")
  ctz-draw("C", stroke: purple)

  ctz-draw(points: ("A", "B"), labels: (A: "left", B:
"right"))
})
```

**Figure**




Unnamed equivalent:

```
ctz-draw(circle-diameter: ("A", "B"), stroke: purple)
```

## 9.4. Circumcircle — `ctz-def-circumcircle`

Define the circumscribed circle of a triangle (passes through all three vertices).

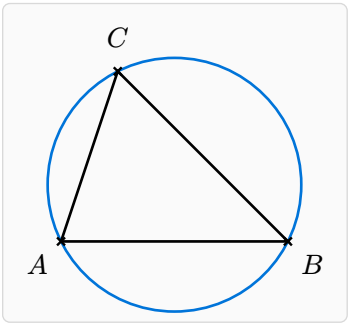
**Code**

```
#ctz-canvas(length: 0.7cm, {
  import cetz.draw: *
  ctz-init()
  ctz-def-points(A: (0, 0), B: (4, 0), C: (1, 3))

  ctz-def-circumcircle("circ", "A", "B", "C")
  ctz-draw("circ", stroke: blue)
  ctz-draw(line: ("A", "B", "C", "A"), stroke: black)

  ctz-draw(points: ("A", "B", "C"), labels: (
    A: "below left", B: "below right", C: "above"))
})
```

**Figure**




Unnamed equivalent:

```
ctz-draw(circumcircle: ("A", "B", "C"), stroke: blue)
```

## 9.5. Incircle — `ctz-def-incircle`

Define the inscribed circle of a triangle (tangent to all three sides).

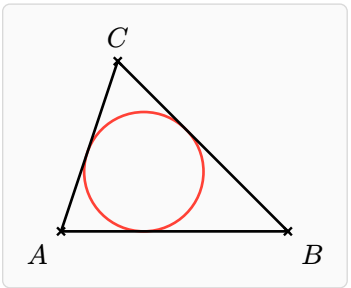
**Code**

```
#ctz-canvas(length: 0.7cm, {
  import cetz.draw: *
  ctz-init()
  ctz-def-points(A: (0, 0), B: (4, 0), C: (1, 3))

  ctz-def-incircle("inc", "A", "B", "C")
  ctz-draw("inc", stroke: red)
  ctz-draw(line: ("A", "B", "C", "A"), stroke: black)

  ctz-draw(points: ("A", "B", "C"), labels: (
    A: "below left", B: "below right", C: "above"))
})
```

**Figure**




Unnamed equivalent:

```
ctz-draw(incircle: ("A", "B", "C"), stroke: red)
```

## 9.6. Excircle — `ctz-def-excircle`

Define an excircle of a triangle (tangent to one side and the extensions of the other two sides). The `vertex` parameter specifies which vertex the excircle is opposite to.

**Code**

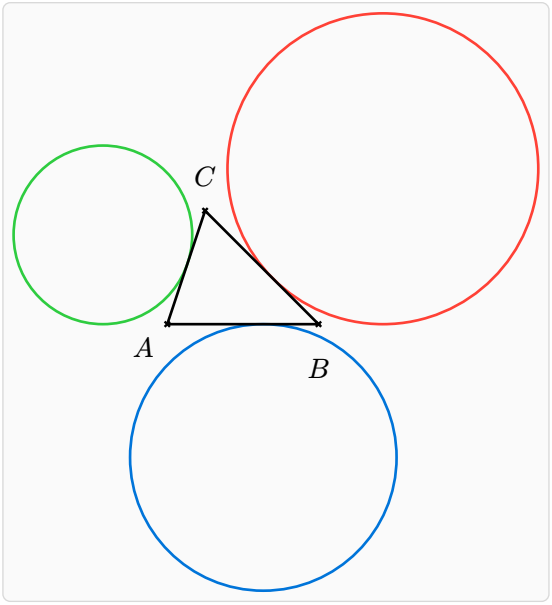
```
#ctz-canvas(length: 0.5cm, {
  import cetz.draw: *
  ctz-init()
  ctz-def-points(A: (0, 0), B: (4, 0), C: (1, 3))

  ctz-def-excircle("excA", "A", "B", "C", vertex:
"a")
  ctz-def-excircle("excB", "A", "B", "C", vertex:
"b")
  ctz-def-excircle("excC", "A", "B", "C", vertex:
"c")

  ctz-draw("excA", stroke: red)
  ctz-draw("excB", stroke: green)
  ctz-draw("excC", stroke: blue)
  ctz-draw(line: ("A", "B", "C", "A"), stroke: black)

  ctz-draw(points: ("A", "B", "C"), labels: (
    A: "below left", B: "below right", C: "above"))
})
```

**Figure**




## 9.7. Nine-Point (Euler) Circle — `ctz-def-euler-circle`

Define the nine-point circle of a triangle. This circle passes through nine significant points: the midpoints of the three sides, the feet of the three altitudes, and the midpoints of the segments from vertices to the orthocenter.

**Code**

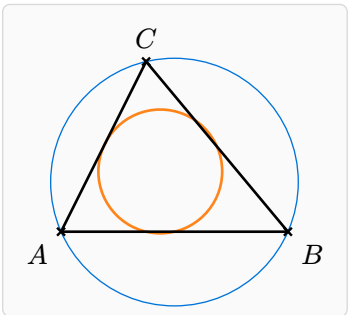
```
#ctz-canvas(length: 0.7cm, {
  import cetz.draw: *
  ctz-init()
  ctz-def-points(A: (0, 0), B: (4, 0), C: (1.5, 3))

  ctz-def-euler-circle("euler", "A", "B", "C")
  ctz-def-circumcircle("circ", "A", "B", "C")

  ctz-draw("circ", stroke: blue + 0.5pt)
  ctz-draw("euler", stroke: orange + 1pt)
  ctz-draw(line: ("A", "B", "C", "A"), stroke: black)

  ctz-draw(points: ("A", "B", "C"), labels: (
    A: "below left", B: "below right", C: "above"))
})
```

**Figure**




The nine-point circle has radius equal to half the circumradius.

## 9.8. Spieker Circle — `ctz-def-spieker-circle`

Define the Spieker circle of a triangle (the incircle of the medial triangle).

**Code**

```
#ctz-canvas(length: 0.7cm, {
  import cetz.draw: *
  ctz-init()
  ctz-def-points(A: (0, 0), B: (4, 0), C: (1.5, 3))

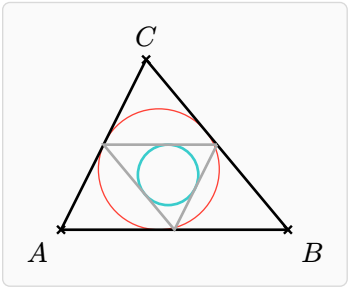
  // Medial triangle
  ctz-def-medial-triangle("Ma", "Mb", "Mc", "A", "B",
"C")

  ctz-def-spieker-circle("spieker", "A", "B", "C")
  ctz-def-incircle("inc", "A", "B", "C")

  ctz-draw("inc", stroke: red + 0.5pt)
  ctz-draw("spieker", stroke: teal + 1pt)
  ctz-draw(line: ("A", "B", "C", "A"), stroke: black)
  ctz-draw(line: ("Ma", "Mb", "Mc", "Ma"), stroke:
gray)

  ctz-draw(points: ("A", "B", "C"), labels: (
    A: "below left", B: "below right", C: "above"))
})
```

**Figure**




## 9.9. Apollonius Circle — `ctz-def-apollonius-circle`

Define an Apollonius circle — the locus of points P such that the ratio PA/PB equals a constant k.

**Code**

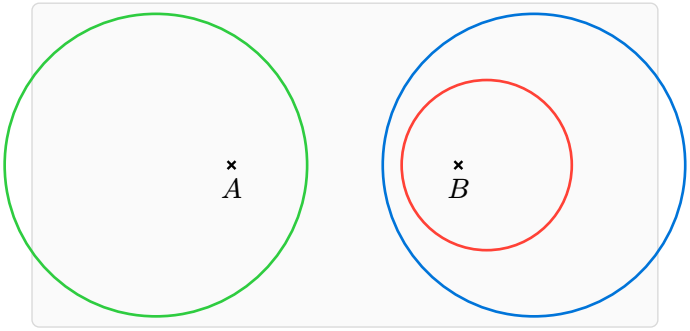
```
#ctz-canvas(length: 0.8cm, {
  import cetz.draw: *
  ctz-init()
  ctz-def-points(A: (-2, 0), B: (2, 0))

  // Circles for different ratios
  ctz-def-apollonius-circle("ap2", "A", "B", 2)
  ctz-def-apollonius-circle("ap3", "A", "B", 3)
  ctz-def-apollonius-circle("ap05", "A", "B", 0.5)

  ctz-draw("ap2", stroke: blue)
  ctz-draw("ap3", stroke: red)
  ctz-draw("ap05", stroke: green)

  ctz-draw(points: ("A", "B"), labels: (A: "below",
B: "below"))
})
```

**Figure**




Note: When k = 1, the locus is the perpendicular bisector of AB (a line, not a circle).

## 9.10. Circle Labels — `ctz-label-circle`

Place a label relative to a named circle:

```
ctz-def-circle("C1", "O", radius: 2)
ctz-draw("C1", stroke: blue)
ctz-label-circle("C1", $C_1$, pos: "above right", dist: 0.2)
```

Parameters:

- `pos`: Position around the circle (`"above"`, `"below"`, `"left"`, `"right"`, or combinations)
- `dist`: Distance from the circle edge
- `offset`: Additional (x, y) offset

## 9.11. Summary: Circle Functions

| Define Function | Draw Shorthand | Description |
|---|---|---|
| `ctz-def-circle(name, center, radius: r)` | `circle-r: (center, r)` | Circle by center and radius |
| `ctz-def-circle(name, center, through: pt)` | `circle-through: (center, pt)` | Circle through point |
| `ctz-def-circle-diameter(name, a, b)` | `circle-diameter: (a, b)` | Circle by diameter |
| `ctz-def-circumcircle(name, a, b, c)` | `circumcircle: (a, b, c)` | Circumcircle of triangle |
| `ctz-def-incircle(name, a, b, c)` | `incircle: (a, b, c)` | Incircle of triangle |
| `ctz-def-excircle(name, a, b, c, vertex: v)` | — | Excircle opposite to vertex |
| `ctz-def-euler-circle(name, a, b, c)` | — | Nine-point circle |
| `ctz-def-spieker-circle(name, a, b, c)` | — | Spieker circle |
| `ctz-def-apollonius-circle(name, a, b, k)` | — | Apollonius circle (ratio k) |

All defined circles can be:

- Drawn with `ctz-draw("name", stroke: ..., fill: ...)`
- Used in circle-circle intersections: `ctz-def-cc(("P", "Q"), "circle1", "circle2")`
- Used in line-circle intersections: `ctz-def-lc(("P", "Q"), line, "circle")`

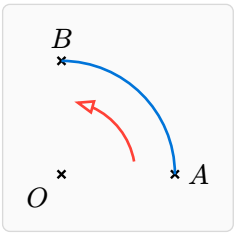
## 9.12. Arcs, Sectors and Semicircles

Arcs are drawn around a **center**, from a start point to an end point (counter-clockwise). The radius is the distance from the center to the start point.

**Code**

```
#ctz-canvas({
  ctz-init()
  ctz-def-points(O: (0, 0), A: (2, 0), B: (0, 2))
  ctz-draw(points: ("O", "A", "B"), labels: true)
  ctz-draw-arc("O", "A", "B", stroke: blue + 1pt)
  ctz-draw-arc-r((0, 0), 1.3, 10deg, 80deg, stroke:
red, arrows: "->")
})
```

**Figure**




**Parameters — `ctz-draw-arc(center, start, end, ..)`:**

- `center`, `start`, `end` — point names or coordinates; the arc runs from `start` to `end` around `center`
- `stroke` — stroke style (default: `black`)
- `arrows` — `"->"`, `"<-"`, `"<->"` or `none`
- `delta` — extend the arc by this many degrees on both sides

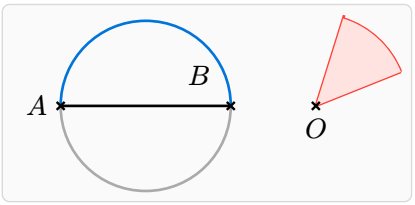
**Parameters — `ctz-draw-arc-r(center, radius, start-angle, end-angle, ..)`:**

- `radius` — number (canvas units)
- `start-angle`, `end-angle` — Typst angles (e.g. `30deg`)

**Code**

```
#ctz-canvas({
  ctz-init()
  ctz-def-points(A: (0, 0), B: (3, 0), O: (4.5, 0),
P: (6, 0.6), Q: (5, 1.6))
  ctz-draw-segment("A", "B")
  ctz-draw-semicircle("A", "B", stroke: blue)
  ctz-draw-semicircle("A", "B", stroke: gray, above:
false)
  ctz-draw-sector("O", "P", "Q", stroke: red, fill:
red.lighten(85%))
  ctz-draw(points: ("A", "B", "O"), labels: true)
})
```

**Figure**




**Parameters — `ctz-draw-semicircle(a, b, ..)`:** `above: true` picks the side of the diameter AB. **`ctz-draw-sector(center, start, end, ..)`** accepts `stroke` and `fill`.

# 10. Clipping

Lines that extend infinitely need to be clipped to the visible region.

## 10.1. Clip to Canvas — `ctz-canvas(clip-canvas: ...)`

You can have the canvas set the clip region automatically by providing a clip rectangle to `ctz-canvas`. This also fixes the canvas bounds to that rectangle:

```
#ctz-canvas(clip-canvas: (-1, -1, 4, 5), {
  import cetz.draw: *
  ctz-init()

  ctz-def-points(A: (0, 0), B: (2, 3))
  ctz-draw-line-add("A", "B", add: (5, 5), stroke: blue)
  ctz-show-clip(stroke: gray + 0.5pt)
})
```

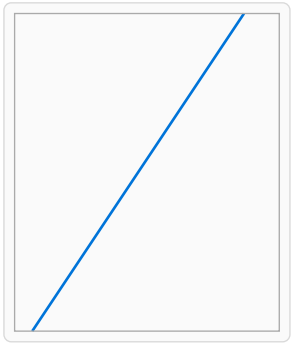
Use `clip-canvas: false` to disable this per-canvas. You can also set a global default with `ctz-default-clip-canvas`.

```
#let ctz-default-clip-canvas = (-1, -1, 4, 5)
#ctz-canvas({
  import cetz.draw: *
  ctz-init()
  // ... drawing code ...
})
```

**Code**

```
#ctz-canvas(length: 0.7cm, clip-canvas: (-1, -1, 4,
5), {
  import cetz.draw: *
  ctz-init()
  ctz-def-points(A: (0, 0), B: (2, 3))
  ctz-draw-line-add("A", "B", add: (5, 5), stroke:
blue)
  ctz-show-clip(stroke: gray + 0.5pt)
})
```

**Figure**



When `clip-canvas` is set, **only line-based elements** are clipped to the canvas bounds (extended lines and segments). Circles, arcs, and other shapes are not clipped yet. You can use the standard drawing helpers for lines:

```
ctz-draw-line-add("A", "B", add: (2, 2), stroke: blue)
ctz-draw-segment("A", "B", stroke: red)
```

# 11. Grid & Annotation Helpers

This section covers helper functions for creating structured diagrams with grids, highlights, and annotations. These are particularly useful for educational materials like Pascal's triangle, multiplication tables, or any grid-based visualizations.

## 11.1. Grid Positioning

Three grid layout systems are provided for positioning content.

### 11.1.1. Triangular Grid

The `triangular-pos(row, col)` function computes positions in a triangular layout where each row has one more element than the previous, centered horizontally. This is the layout used by Pascal's triangle.

**Code**

```
#cetz.canvas({
  import cetz.draw: *

  draw-triangular-grid(
    cetz.draw, 5,
    (row, col) => str(row) + "," + str(col),
    h-spacing: 1.8,
    v-spacing: 1.0,
    text-size: 9pt,
  )
})
```

**Figure**




**Parameters:**

- `row` — Row number (0 = top)
- `col` — Column within row (0 to row)
- `h-spacing` — Horizontal spacing between adjacent cells (default: 1.5)
- `v-spacing` — Vertical spacing between rows (default: 1.2)
- `origin` — Grid origin point (default: (0, 0))

### 11.1.2. Pascal's Triangle

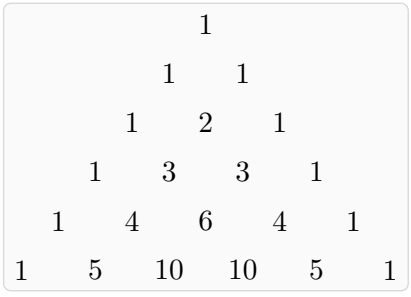
For Pascal's triangle binomial coefficients, use `draw-pascal-values`:

**Code**

```
#cetz.canvas({
  import cetz.draw: *
  import "@preview/ctz-euclide:0.2.1": *

  draw-pascal-values(cetz.draw, 6,
    h-spacing: 1.5, v-spacing: 1.0,
    text-size: 11pt)
})
```

**Figure**



### 11.1.3. Row and Diagonal Labels

Add row labels (`draw-row-labels`) and diagonal labels (`draw-diagonal-labels`):

**Code**

```
#cetz.canvas({
  import cetz.draw: *

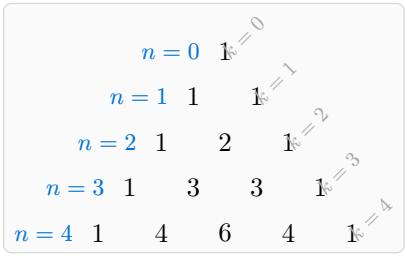
  let sp = (h: 1.4, v: 1.0)

  draw-pascal-values(cetz.draw, 5,
    h-spacing: sp.h, v-spacing: sp.v,
    text-size: 10pt)

  draw-row-labels(cetz.draw, 5,
    h-spacing: sp.h, v-spacing: sp.v,
    offset: -1.2, text-color: blue,
    format: n => $n=#n$)

  draw-diagonal-labels(cetz.draw, 5,
    h-spacing: sp.h, v-spacing: sp.v,
    offset: (0.4, 0.3), text-color: gray,
    format: k => $k=#k$)
})
```

**Figure**




### 11.1.4. Rectangular Grid

The `grid-pos(row, col)` function computes positions in a standard rectangular grid:

**Code**

```
#cetz.canvas({
  import cetz.draw: *
  import "@preview/ctz-euclide:0.2.1": *

  draw-rectangular-grid(
    cetz.draw, 3, 4,
    (r, c) => str(r * 4 + c + 1),
    h-spacing: 1.2, v-spacing: 1.0,
    text-size: 11pt)
})
```

**Figure**




### 11.1.5. Hexagonal Grid

The `hex-pos(row, col)` function computes positions in a hexagonal (honeycomb) grid:

**Code**

```
#cetz.canvas({
  import cetz.draw: *

  for row in range(3) {
    for col in range(4) {
      let p = hex-pos(row, col, size: 0.8)
      cetz.draw.circle(p, radius: 0.35,
        stroke: blue + 0.5pt,
        fill: blue.lighten(90%))
      content(p, text(size: 8pt,
        str(row) + "," + str(col)))
    }
  }
})
```

**Figure**




## 11.2. Annotation Helpers

Functions for highlighting cells and drawing annotation arrows.

### 11.2.1. Cell Highlighting

`highlight-fill` draws a filled circle, `highlight-outline` draws an outlined circle, and `highlight-many` highlights multiple positions:

**Code**

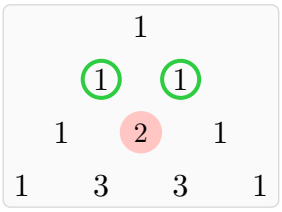
```
#cetz.canvas({
  import cetz.draw: *
  import "@preview/ctz-euclide:0.2.1": *

  draw-pascal-values(cetz.draw, 4,
    h-spacing: 1.5, v-spacing: 1.0)

  // Filled highlight
  let p1 = triangular-pos(2, 1,
    h-spacing: 1.5, v-spacing: 1.0)
  highlight-fill(cetz.draw, p1,
    radius: 0.4, fill: red.lighten(70%))
  content(p1, "2")

  // Outline highlights
  let p2 = triangular-pos(1, 0,
    h-spacing: 1.5, v-spacing: 1.0)
  let p3 = triangular-pos(1, 1,
    h-spacing: 1.5, v-spacing: 1.0)
  highlight-many(cetz.draw, (p2, p3),
    radius: 0.35, stroke: green + 1.5pt)
})
```

**Figure**




### 11.2.2. Curved Arrows

`curved-arrow` draws a curved annotation arrow. The `bend` parameter controls curvature direction and amount (positive = bend right, negative = bend left):

**Code**

```
#cetz.canvas({
  import cetz.draw: *

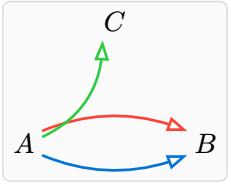
  content((0, 0), $A$)
  content((3, 0), $B$)
  content((1.5, 2), $C$)

  curved-arrow(cetz.draw,
    (0.3, 0.2), (2.7, 0.2),
    bend: 0.5, stroke: red + 1pt)

  curved-arrow(cetz.draw,
    (0.3, -0.2), (2.7, -0.2),
    bend: -0.5, stroke: blue + 1pt)

  curved-arrow(cetz.draw,
    (0.3, 0.1), (1.3, 1.7),
    bend: -0.6, stroke: green + 1pt)
})
```

**Figure**




### 11.2.3. Smooth Arrows

`smooth-arrow` draws a Catmull-Rom spline through waypoints for complex curved paths:

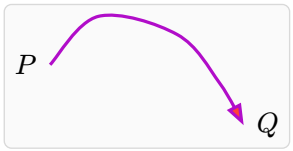
**Code**

```
#cetz.canvas({
  import cetz.draw: *
  import "@preview/ctz-euclide:0.2.1": *

  content((0, 0), $P$)
  content((4, -1), $Q$)

  smooth-arrow(cetz.draw,
    (0.4, 0), (3.6, -1),
    waypoints: (
      (1.2, 0.8),
      (2.5, 0.5),
      (3.2, -0.3),
    ),
    stroke: purple + 1.2pt)
})
```

**Figure**




### 11.2.4. Addition Indicator

`draw-addition-indicator` shows two values combining into a result (useful for Pascal's triangle):

**Code**

```
#cetz.canvas({
  import cetz.draw: *

  let a = (0, 0)
  let b = (1.5, 0)
  let sum = (0.75, -1.2)

  content(a, $3$)
  content(b, $3$)
  content(sum, $6$)

  draw-addition-indicator(cetz.draw,
    a, b, sum, color: green)
})
```

**Figure**




### 11.2.5. Braces

`draw-brace-h` and `draw-brace-v` draw horizontal and vertical braces with optional labels:

**Code**

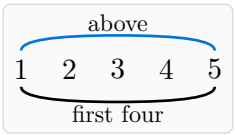
```
#cetz.canvas({
  import cetz.draw: *
  import "@preview/ctz-euclide:0.2.1": *

  for i in range(5) {
    content((i * 0.8, 0),
      text(size: 11pt, str(i + 1)))
  }

  draw-brace-h(cetz.draw,
    (0, -0.3), (3.2, -0.3),
    label: text(size: 9pt, "first four"),
    side: "below", amplitude: 0.25)

  draw-brace-h(cetz.draw,
    (0, 0.3), (3.2, 0.3),
    label: text(size: 9pt, "above"),
    side: "above", amplitude: 0.25,
    stroke: blue + 1pt)
})
```

**Figure**



**Code**

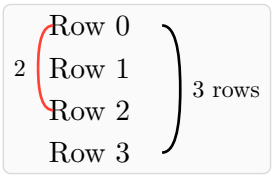
```
#cetz.canvas({
  import cetz.draw: *

  for i in range(4) {
    content((0, -i * 0.7),
      text(size: 11pt, "Row " + str(i)))
  }

  draw-brace-v(cetz.draw,
    (1.2, 0), (1.2, -2.1),
    label: text(size: 9pt, "3 rows"),
    side: "right", amplitude: 0.3)

  draw-brace-v(cetz.draw,
    (-0.6, 0), (-0.6, -1.4),
    label: text(size: 9pt, "2"),
    side: "left", amplitude: 0.25,
    stroke: red + 1pt)
})
```

**Figure**




## 12. Raw Algorithms

For direct computation without the point registry, use the `raw` dictionary:

```
// Direct centroid calculation
let center = raw.ctz-def-centroid((0,0,0), (3,0,0), (1.5,2.5,0))

// Distance between points
let d = raw.dist((0,0,0), (3,4,0))  // Returns 5

// Line-line intersection
let pt = raw.line-line((0,0,0), (1,1,0), (0,1,0), (1,0,0), ray: true)
```

Available raw functions:

- **Intersections**: `line-line`, `line-circle`, `circle-circle`
- **Triangle centers**: `ctz-def-centroid`, `ctz-def-circumcenter`, `ctz-def-incenter`, `ctz-def-orthocenter`, `euler-center`, `ctz-def-lemoine`, etc.
- **Transformations**: `ctz-def-rotation`, `reflection`, `translation`, `ctz-def-homothety`, `projection`, `ctz-def-inversion`
- **Utilities**: `ctz-def-midpoint`, `dist`, `angle-at-vertex`, `triangle-area`, etc.

### 12.1. Measurements — `ctz-calc-*`

These return values through the canvas context (use them inside a canvas via `ctz-algorithms` for direct numeric work instead):

- `ctz-calc-length(a, b)` — distance between two named points
- `ctz-calc-angle(a, b, c)` — angle at vertex `b`, in degrees
- `ctz-calc-area(a, b, c)` — triangle area
- `ctz-calc-slope(a, b)` — slope angle of the line ab, in degrees
- `ctz-calc-ratio(a, b, c)` — ratio AB/AC

For direct computation on raw coordinates, prefer the `ctz-algorithms` dictionary (call entries with the parenthesized form):

```
#let d = (ctz-algorithms.dist)((0, 0), (3, 4, 0))   // 5
#let s = (ctz-algorithms.slope-angle)((0, 0), (1, 1))  // 45
#let I = (ctz-algorithms.incenter)(A, B, C)
```

### 12.2. Axes and Guide Lines

- `ctz-draw-axes(xmin: .., xmax: .., ymin: .., ymax: .., ..)` — both axes with ticks, numeric labels and arrowheads
- `ctz-draw-axis-x(..)` / `ctz-draw-axis-y(..)` — one axis only
- `ctz-draw-hline(y, xmin: .., xmax: .., ..)` — horizontal guide line
- `ctz-draw-vline(x, ymin: .., ymax: .., ..)` — vertical guide line
- `ctz-draw-text(x, y, content, angle: 0deg, anchor: "center")` — free text

## 13. Function Reference

### 13.1. Point Definition

- `ctz-def-points(A: (x, y), ...)` — Define named points
- `ctz-def-line(name, a, b)` — Named line from two points
- `ctz-def-circle(name, center, radius|through)` — Named circle
- `ctz-def-polygon(name, a, b, c, ...)` — Named polygon
- `ctz-def-midpoint(name, a, b)` — Midpoint of segment
- `ctz-def-linear(name, a, b, k)` — Point at ratio k along line
- `ctz-def-regular-polygon([name,] names, center, first)` — Regular n-gon vertices (optionally register polygon name)
- `ctz-def-point-on-circle(name, center, radius, angle)` — Point on circle at angle
- `ctz-def-equilateral(name, a, b)` — Third vertex of equilateral triangle
- `ctz-def-golden(name, a, b, inside: true)` — Golden ratio point (`inside: false` for the external section)
- `ctz-def-square(nc, nd, a, b)` — Third and fourth vertices of square ABCD
- `ctz-def-barycentric(name, a, b, c, wa, wb, wc)` — Barycentric combination point
- `ctz-def-symmetry(name, source, center)` — Central symmetry image
- `ctz-def-tangent-from(n1, n2, point, center, radius)` — Tangent points from external point to circle
- `ctz-def-radical-axis(n1, n2, circle1, circle2)` — Two points on the radical axis of two circles
- `ctz-pt(name)` / `ctz-pts(..)` — Aliases: point-name coordinate helper / `ctz-def-points`

### 13.2. Line Constructions

- `ctz-def-perp(n1, n2, line, through)` — Perpendicular through point
- `ctz-def-para(n1, n2, line, through)` — Parallel through point
- `ctz-def-bisect(n1, n2, a, vertex, c)` — Angle bisector
- `ctz-def-mediator(n1, n2, a, b)` — Perpendicular bisector

### 13.3. Intersections

- `ctz-def-ll(name, line1, line2)` — Line-line intersection

- `ctz-def-lc(names, line, circle)` — Line-circle intersections
- `ctz-def-cc(names, circle1, circle2)` — Circle-circle intersections

### 13.4. Triangle Centers

- `ctz-def-centroid`, `ctz-def-circumcenter`, `ctz-def-incenter`, `ctz-def-orthocenter`
- `ctz-def-euler`, `ctz-def-lemoine`, `ctz-def-nagel`, `ctz-def-gergonne`, `ctz-def-spieker`
- `ctz-def-feuerbach`, `ctz-def-mittenpunkt`, `ctz-def-excenter`

### 13.5. Special Triangles

- `ctz-def-medial-triangle(na, nb, nc, a, b, c)` — Medial triangle
- `ctz-def-orthic-triangle(na, nb, nc, a, b, c)` — Orthic triangle
- `ctz-def-intouch-triangle(na, nb, nc, a, b, c)` — Intouch triangle
- `ctz-def-thales-triangle(na, nb, nc, center, radius, ...)` — Right triangle via Thales' theorem

### 13.6. Transformations

- `ctz-def-rotation(name, source, center, angle)` — Rotation (works on all object types)
- `ctz-def-reflect(name, source, line-a, line-b)` — Reflection
- `ctz-def-translate(name, source, vector)` — Translation
- `ctz-def-homothety(name, source, center, factor)` — Homothety
- `ctz-def-project(name, source, line-a, line-b)` — Projection
- `ctz-def-inversion(name, source, center, radius)` — Inversion (works on points, lines, circles, polygons)
- `ctz-duplicate(target, source, points: auto)` — Duplicate any geometric object

### 13.7. Drawing

- `ctz-draw(name, ...)` or `ctz-draw(points: ..., labels: ...)` — Draw any object type (polymorphic) or draw/label points using unified API
- `ctz-draw-points(names...)` — Draw points (legacy, prefer unified `ctz-draw()`)
- `ctz-draw-labels(names, placements)` — Label points (legacy, prefer unified `ctz-draw()`)
- `ctz-style(point: (...))` — Set styling
- `ctz-draw-angle(vertex, a, b, ...)` — Mark angle
- `ctz-draw-mark-right-angle(a, vertex, c, ...)` — Right angle mark
- `ctz-draw-segment(a, b, ...)` — Draw segment
- `ctz-draw-measure-segment(a, b, ...)` — Offset measurement with fences and label
- `ctz-draw-path(spec, ...)` — Draw path with per-segment tips
- `ctz-draw-polygon(points...)` — Draw polygon (triangle, quadrilateral, etc.)
- `ctz-draw-fill-polygon(points...)` — Fill polygon
- `ctz-draw-regular-polygon(names, ...)` — Draw regular polygon by vertex names
- `ctz-draw-fill-regular-polygon(names, ...)` — Fill regular polygon by vertex names
- `ctz-draw-circle(name, ...)` — Draw named circle
- `ctz-label-circle(name, label, ...)` — Label named circle
- `ctz-label-polygon(name, label, ...)` — Label named polygon
- `ctz-draw-circumcircle(a, b, c, ...)` — Circumscribed circle
- `ctz-draw-incircle(a, b, c, ...)` — Inscribed circle
- `ctz-draw-circle-r(center, radius, ...)` — Circle by center and radius
- `ctz-draw-circle-through(center, through, ...)` — Circle by center and a point on it
- `ctz-draw-circle-diameter(a, b, ...)` — Circle by diameter endpoints
- `ctz-draw-semicircle(a, b, above: true, ...)` — Semicircle on diameter AB
- `ctz-draw-arc(center, start, end, ...)` — Arc around a center between two points
- `ctz-draw-arc-r(center, radius, start-angle, end-angle, ...)` — Arc by radius and angles
- `ctz-draw-sector(center, start, end, fill: ..., ...)` — Circular sector
- `ctz-draw-ellipse(center, rx, ry, angle: ..., ...)` — Ellipse
- `ctz-draw-projectile(origin, velocity, ...)` — Projectile trajectory
- `ctz-draw-ellipse-tangent(center, rx, ry, t, ...)` — Tangent to ellipse at parameter t

- `ctz-draw-parabola-tangent(focus, directrix, t, ...)` — Tangent to parabola at parameter t
- `ctz-draw-parabola-tangents-from-point(focus, directrix, p, ...)` — Parabola tangents from a point
- `ctz-draw-label-segment(a, b, label: ..., ...)` — Text label on a segment
- `ctz-draw-label-angle(vertex, a, c, label: ..., ...)` — Text label inside an angle
- `ctz-draw-text(x, y, content, ...)` — Free text at coordinates
- `ctz-draw-hline(y, ...)` / `ctz-draw-vline(x, ...)` — Horizontal/vertical guide line
- `ctz-draw-axis-x(...)` / `ctz-draw-axis-y(...)` — Single axis with ticks and labels
- `ctz-points(..)` — Alias of `ctz-draw-points`
- `ctz-label-avoid(setting)` — Enable/disable/configure automatic label placement
- `ctz-main-levee(enabled: true, roughness: ..., seed: ...)` — Toggle hand-drawn mode globally

## 13.8. Clipping

- `ctz-canvas(clip-canvas: ...)` — Set clip bounds for the canvas
- `ctz-set-clip(xmin, ymin, xmax, ymax)` / `ctz-clear-clip()` — Set/clear the global clip region
- `ctz-show-clip(...)` — Draw the clip boundary
- `ctz-draw-line-add(a, b, ...)` — Extended line (respects clip bounds)
- `ctz-draw-segment(a, b, ...)` — Segment (respects clip bounds)
- `ctz-draw-line-global-clip(a, b, ...)` / `ctz-draw-seg-global-clip(a, b, ...)` — Explicitly clipped line/segment
- `ctz-draw-clipped-line(p1, p2, bounds...)` / `ctz-draw-clipped-line-add(a, b, bounds..., add: ...)` — Clip to manual bounds
- `ctz-draw-clip-boundary(bounds..., ...)` — Draw a manual clip rectangle
- `ctz-clip-line-to-rect(x0, y0, x1, y1, bounds...)` — Raw Cohen–Sutherland clipping (returns endpoints or `none`)

## 13.9. Calculations & Raw Utilities

- `ctz-calc-length(a, b)` / `ctz-calc-angle(a, b, c)` / `ctz-calc-area(a, b, c)` / `ctz-calc-slope(a, b)` / `ctz-calc-ratio(a, b, c)` — Measurements returned via the canvas context
- `ctz-algorithms` — Dictionary of all raw (context-free) algorithms; call as `(ctz-algorithms.dist)(p, q)`
- `ctz-dist(a, b)` — Distance between raw coordinates
- `ctz-angle-to(a, b)` — Direction angle from a to b
- `ctz-midpoint-raw(a, b)` — Midpoint of raw coordinates
- `ctz-project-point-on-line(p, a, b)` — Foot of the perpendicular from p onto line ab
- `ctz-rotate-point(p, center, angle)` — Rotate a raw coordinate
- `ctz-version` — Package version as an (major, minor, patch) array

## 13.10. Style Constants

Read-only defaults, overridable per canvas via `ctz-style(..)`: `ctz-default-point-shape`, `ctz-default-point-size`, `ctz-default-point-size-pt`, `ctz-default-point-size-small-pt`, `ctz-default-point-color`, `ctz-default-point-stroke-width`, `ctz-default-line-color`, `ctz-default-mark-color`, `ctz-default-mark-stroke-width`, `ctz-default-main-levee`, `ctz-default-main-levee-roughness`, `ctz-default-main-levee-seed`, `ctz-default-clip-canvas`.

## 13.11. Advanced Ellipse Constructions

- `ctz-def-ellipse-project(name, center, rx, ry, point, ...)` — Project a point onto an ellipse
- `ctz-ellipse-tangency-geom(a, b)` / `ctz-draw-ellipse-tangency(a, b)` — Full ellipse/circles tangency construction
- `ctz-ellipse-tangent-from-point-towards(...)` / `ctz-draw-ellipse-tangent-from-point-towards(...)` — Directed tangent selection

## 13.12. Grid Positioning

- `triangular-pos(row, col, ...)` — Position in triangular grid (Pascal layout)
- `grid-pos(row, col, ...)` — Position in rectangular grid

- `hex-pos(row, col, ...)` — Position in hexagonal grid
- `binomial(n, k)` — Binomial coefficient C(n,k)

## 13.13. Grid Drawing

- `draw-triangular-grid(cetz-draw, rows, content-fn, ...)` — Draw triangular grid with custom content
- `draw-rectangular-grid(cetz-draw, rows, cols, content-fn, ...)` — Draw rectangular grid
- `draw-pascal-values(cetz-draw, rows, ...)` — Draw Pascal's triangle values
- `draw-row-labels(cetz-draw, rows, ...)` — Draw row labels
- `draw-diagonal-labels(cetz-draw, count, ...)` — Draw diagonal labels

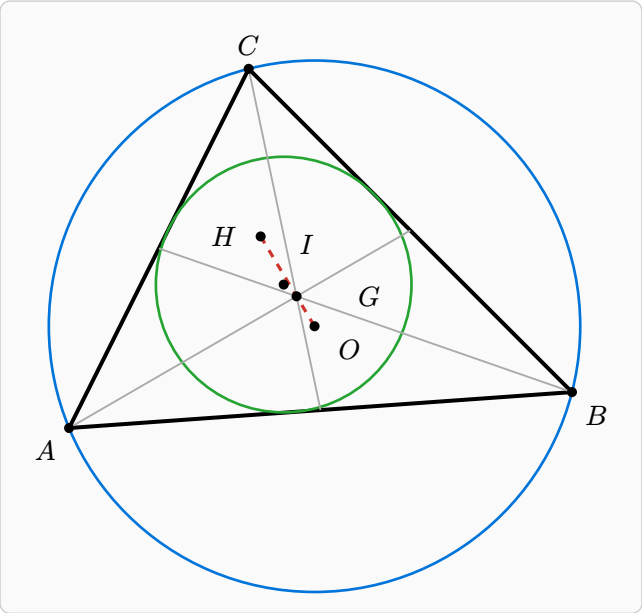
## 13.14. Annotations

- `highlight-fill(cetz-draw, pos, ...)` — Filled circle highlight
- `highlight-outline(cetz-draw, pos, ...)` — Outlined circle highlight
- `highlight-many(cetz-draw, positions, ...)` — Highlight multiple positions
- `curved-arrow(cetz-draw, from, to, ...)` — Curved annotation arrow
- `smooth-arrow(cetz-draw, from, to, waypoints, ...)` — Smooth spline arrow
- `draw-addition-indicator(cetz-draw, a, b, result, ...)` — Show addition relationship
- `draw-brace-h(cetz-draw, start, end, ...)` — Horizontal brace
- `draw-brace-v(cetz-draw, start, end, ...)` — Vertical brace

# 14. Gallery Examples

The following pages showcase advanced geometric constructions using ctz-euclide. Each example demonstrates different features and techniques.

### Triangle Centers




```
#ctz-canvas(length: 0.95cm, {
  import cetz.draw: *
  ctz-init()
  ctz-style(point: (shape: "dot", size: 0.07, fill: black))

  ctz-def-points(A: (0, 0), B: (7, 0.5), C: (2.5, 5))
  ctz-draw(line: ("A", "B", "C", "A"), stroke: black + 1.5pt)

  ctz-def-centroid("G", "A", "B", "C")
  ctz-def-circumcenter("O", "A", "B", "C")
  ctz-def-incenter("I", "A", "B", "C")
  ctz-def-orthocenter("H", "A", "B", "C")

  // Euler line (H, G, O are collinear)
  ctz-draw(segment: ("H", "O"), stroke: (
    paint: red.darken(20%),
    dash: "dashed",
    thickness: 1.2pt
  ))

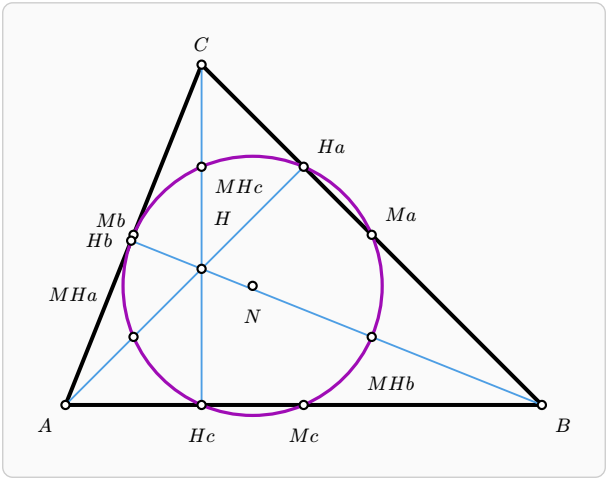
  // Circumcircle and incircle
  ctz-draw(circle-through: ("O", "A"), stroke: blue + 1pt)
  ctz-draw(incircle: ("A", "B", "C"), stroke: green.darken(20%) + 1pt)

  // Draw medians to centroid
  ctz-def-midpoint("Ma", "B", "C")
  ctz-def-midpoint("Mb", "A", "C")
  ctz-def-midpoint("Mc", "A", "B")
  ctz-draw(segment: ("A", "Ma"), stroke: gray + 0.7pt)
  ctz-draw(segment: ("B", "Mb"), stroke: gray + 0.7pt)
  ctz-draw(segment: ("C", "Mc"), stroke: gray + 0.7pt)

  ctz-draw(points: ("A", "B", "C", "G", "O", "I", "H"), labels: (
    A: "below left",
    B: "below right",
    C: "above",
    G: "right",
    O: "below",
    I: "right",
```

```
    H: "left"
  ))
})
```

## Nine-Point Circle




```
#ctz-canvas(length: 0.9cm, {
  import cetz.draw: *
  ctz-init()
  ctz-style(point: (shape: "circle", size: 0.06, stroke: black + 0.8pt, fill: white))

  ctz-def-points(A: (0, 0), B: (7, 0), C: (2, 5))
  ctz-draw(line: ("A", "B", "C", "A"), stroke: black + 1.5pt)

  // Midpoints of sides
  ctz-def-midpoint("Ma", "B", "C")
  ctz-def-midpoint("Mb", "A", "C")
  ctz-def-midpoint("Mc", "A", "B")

  // Feet of altitudes
  ctz-def-project("Ha", "A", "B", "C")
  ctz-def-project("Hb", "B", "A", "C")
  ctz-def-project("Hc", "C", "A", "B")

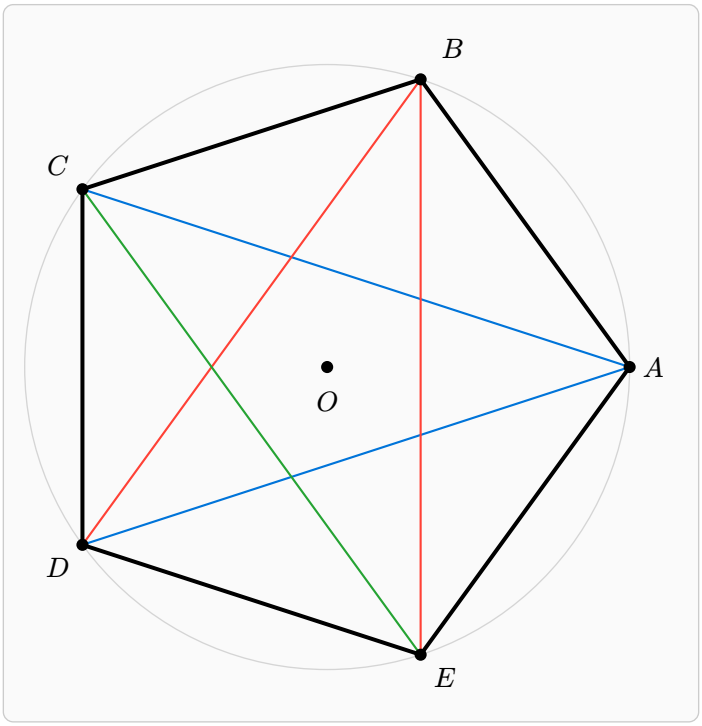
  // Orthocenter and midpoints to vertices
  ctz-def-orthocenter("H", "A", "B", "C")
  ctz-def-midpoint("MHa", "H", "A")
  ctz-def-midpoint("MHb", "H", "B")
  ctz-def-midpoint("MHc", "H", "C")

  // Nine-point circle passes through all 9 points
  ctz-def-euler("N", "A", "B", "C")
  ctz-draw(circle-through: ("N", "Ma"), stroke: purple.darken(10%) + 1.2pt)

  // Draw altitudes
  ctz-draw(segment: ("A", "Ha"), stroke: blue.lighten(30%) + 0.7pt)
  ctz-draw(segment: ("B", "Hb"), stroke: blue.lighten(30%) + 0.7pt)
  ctz-draw(segment: ("C", "Hc"), stroke: blue.lighten(30%) + 0.7pt)

  ctz-draw(points: ("A", "B", "C", "N", "Ma", "Mb", "Mc", "Ha", "Hb", "Hc", "MHa", "MHb", "MHc", "H"), labels: (
    A: "left",
    B: "right",
    C: "above",
    N: "below"
  ))
})
```

## Regular Pentagon




```
#ctz-canvas(length: 1cm, {
  import cetz.draw: *
  ctz-init()
  ctz-style(point: (shape: "dot", size: 0.08, fill: black))

  ctz-def-points(O: (0, 0), V1: (4, 0))
  ctz-def-regular-polygon("Pent", ("A", "B", "C", "D", "E"), "O", "V1")

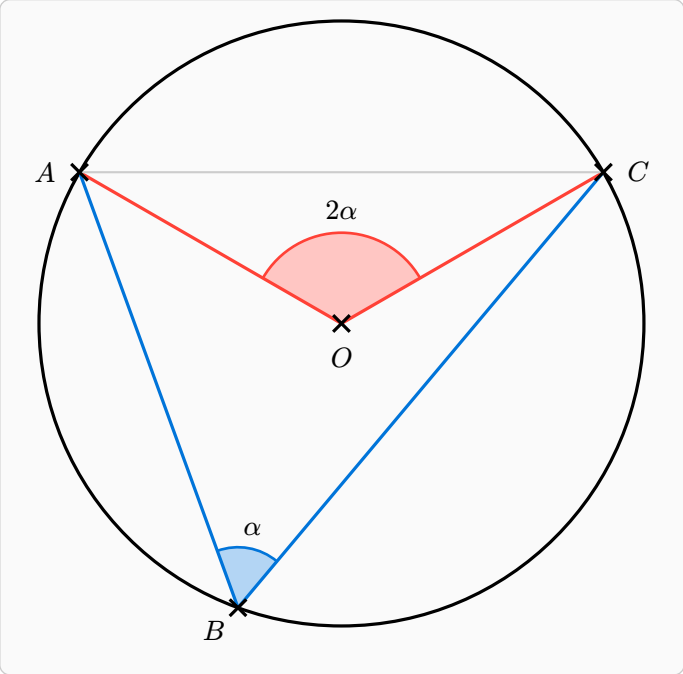
  // Pentagon
  ctz-draw("Pent", stroke: black + 1.5pt)

  // All diagonals
  ctz-draw(segment: ("A", "C"), stroke: blue + 0.8pt)
  ctz-draw(segment: ("A", "D"), stroke: blue + 0.8pt)
  ctz-draw(segment: ("B", "D"), stroke: red + 0.8pt)
  ctz-draw(segment: ("B", "E"), stroke: red + 0.8pt)
  ctz-draw(segment: ("C", "E"), stroke: green.darken(20%) + 0.8pt)

  // Center
  ctz-draw(circle-r: ("O", 4), stroke: gray.lighten(50%) + 0.5pt)

  ctz-draw(points: ("A", "B", "C", "D", "E", "O"), labels: (
    A: "right",
    B: (pos: "above right", offset: (0.1, 0.1),
    C: "above left",
    D: "below left",
    E: "below right",
    O: "below"
  ))
})
```

## Inscribed Angle Theorem




```
#ctz-canvas(length: 1cm, {
  import cetz.draw: *
  ctz-init()
  ctz-style(point: (shape: "cross", size: 0.11, stroke: black + 1.2pt))

  ctz-def-points(O: (0, 0), R: (4, 0))
  ctz-draw(circle-r: ("O", 4), stroke: black + 1.2pt)

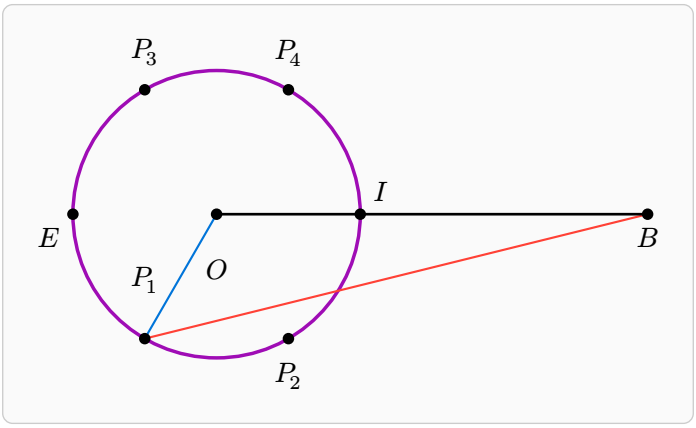
  // Place points on circle
  ctz-def-rotation("A", "R", "O", 150)
  ctz-def-rotation("C", "R", "O", 30)
  ctz-def-rotation("B", "R", "O", 250)

  // Inscribed angle at B
  ctz-draw(segment: ("A", "B"), stroke: blue + 1.2pt)
  ctz-draw(segment: ("B", "C"), stroke: blue + 1.2pt)
  ctz-draw-angle("B", "A", "C", label: $alpha$, radius: 0.8,
    fill: blue.lighten(70%), stroke: blue)

  // Central angle at O (twice the inscribed angle)
  ctz-draw(segment: ("O", "A"), stroke: red + 1.2pt)
  ctz-draw(segment: ("O", "C"), stroke: red + 1.2pt)
  ctz-draw-angle("O", "A", "C", label: $2alpha$, radius: 1.2,
    fill: red.lighten(70%), stroke: red)

  ctz-draw(segment: ("A", "C"), stroke: gray.lighten(40%) + 0.8pt)
  ctz-draw(points: ("O", "A", "B", "C"), labels: (
    O: "below",
    A: "left",
    B: "right",
    C: "above"
  ))
})
```

# Apollonius Circle




```
#ctz-canvas(length: 0.95cm, {
  import cetz.draw: *
  ctz-init()
  ctz-style(point: (shape: "dot", size: 0.08, fill: black))

  ctz-def-points(A: (-3, 0), B: (3, 0))

  // Apollonius circle: locus of points P where PA/PB = k
  let k = 2

  // External and internal division points
  ctz-def-points(E: (-5, 0), I: (-1, 0))

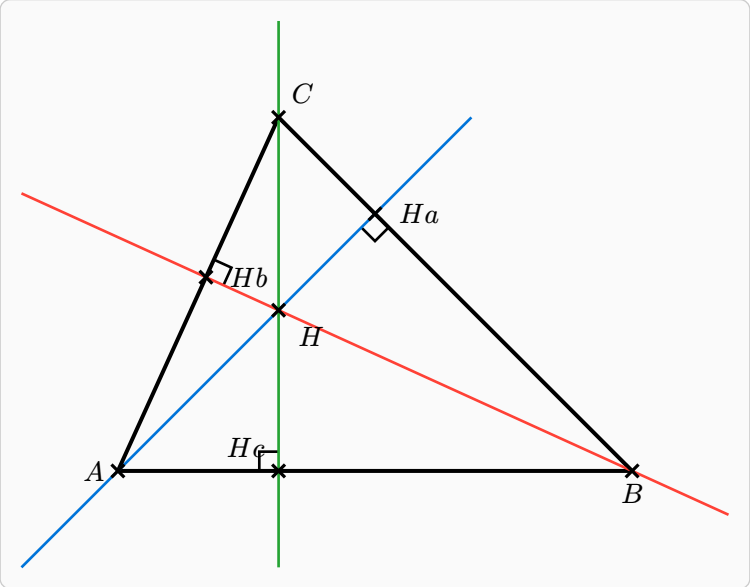
  // Center is midpoint of E and I
  ctz-def-midpoint("O", "E", "I")
  ctz-draw(circle-through: ("O", "E"), stroke: purple.darken(10%) + 1.3pt)

  // Show some points on the circle
  ctz-def-rotation("P1", "E", "O", 60)
  ctz-def-rotation("P2", "E", "O", 120)
  ctz-def-rotation("P3", "E", "O", -60)
  ctz-def-rotation("P4", "E", "O", -120)

  // For P1: PA/PB = k = 2
  ctz-draw(segment: ("P1", "A"), stroke: blue + 0.8pt)
  ctz-draw(segment: ("P1", "B"), stroke: red + 0.8pt)
  ctz-draw(segment: ("A", "B"), stroke: black + 1pt)

  ctz-draw(points: ( "B", "O", "E", "I", "P1", "P2", "P3", "P4"), labels: (
    B: "below right",
    E: "left",
    I: "above right",
    P1: "above",
    O: "below"
  ))
})
```

# Orthocenter and Altitudes




```
#ctz-canvas(length: 0.85cm, clip-canvas: (-1.5, -1.5, 9.5, 7), {
  import cetz.draw: *
  ctz-init()
  ctz-style(point: (shape: "cross", size: 0.1, stroke: black + 1.2pt))

  ctz-def-points(A: (0, 0), B: (8, 0), C: (2.5, 5.5))

  // Triangle
  ctz-draw(line: ("A", "B", "C", "A"), stroke: black + 1.5pt)

  // Extended altitudes (automatically clipped)
  ctz-def-perp("Ha1", "Ha2", ("B", "C"), "A")
  ctz-def-perp("Hb1", "Hb2", ("A", "C"), "B")
  ctz-def-perp("Hc1", "Hc2", ("A", "B"), "C")

  ctz-draw-line-add("A", "Ha1", add: (1, 1.5), stroke: blue + 1pt)
  ctz-draw-line-add("B", "Hb1", add: (1, 1.5), stroke: red + 1pt)
  ctz-draw-line-add("C", "Hc1", add: (1, 2.5), stroke: green.darken(20%) + 1pt)

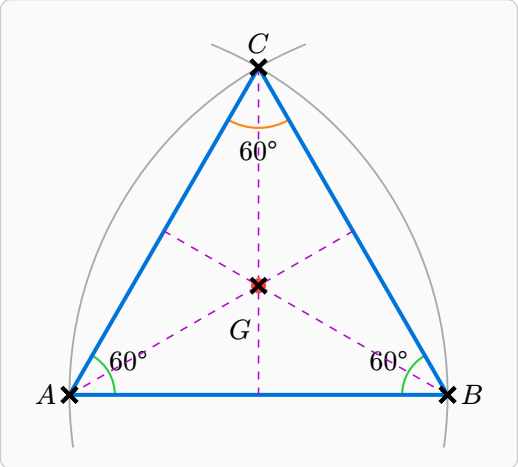
  // Orthocenter (intersection of altitudes)
  ctz-def-orthocenter("H", "A", "B", "C")

  // Feet of altitudes
  ctz-def-project("Ha", "A", "B", "C")
  ctz-def-project("Hb", "B", "A", "C")
  ctz-def-project("Hc", "C", "A", "B")

  ctz-draw-mark-right-angle("A", "Ha", "B", size: 0.3)
  ctz-draw-mark-right-angle("B", "Hb", "C", size: 0.3)
  ctz-draw-mark-right-angle("C", "Hc", "A", size: 0.3)

  ctz-draw(points: ("A", "B", "C", "H", "Ha", "Hb", "Hc"), labels: (
    A: "left",
    B: "below",
    C: "above right",
    H: "right"
  ))
})
```

# Equilateral Triangle Construction




```
#ctz-canvas({
  import cetz.draw: *
  ctz-init()
  ctz-style(point: (shape: "cross", size: 0.1, stroke: black + 1.5pt))

  // Base of equilateral triangle
  ctz-def-points("A", (0, 0), "B", (5, 0))
  ctz-def-equilateral("C", "A", "B")

  // Draw triangle
  ctz-draw(line: ("A", "B", "C", "A"), stroke: blue + 1.5pt)

  // Mark 60° angles
  ctz-draw-angle("A", "B", "C", label: $60°$, radius: 0.6, stroke: green + 0.8pt)
  ctz-draw-angle("B", "C", "A", label: $60°$, radius: 0.6, stroke: green + 0.8pt)
  ctz-draw-angle("C", "A", "B", label: $60°$, radius: 0.8, stroke: orange + 0.8pt)

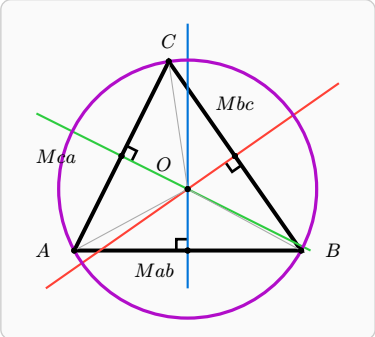
  // Compass construction arcs: C is where the arcs centered at A and B
  // (with radius AB) intersect
  ctz-draw(arc: (center: "A", start: "B", end: "C"), delta: 8, stroke: gray + 0.7pt)
  ctz-draw(arc: (center: "B", start: "C", end: "A"), delta: 8, stroke: gray + 0.7pt)

  // In equilateral triangle, all centers coincide
  ctz-def-centroid("G", "A", "B", "C")

  // Draw medians/altitudes
  ctz-def-midpoint("Ma", "B", "C")
  ctz-def-midpoint("Mb", "A", "C")
  ctz-def-midpoint("Mc", "A", "B")
  ctz-draw(segment: ("A", "Ma"), stroke: (paint: purple, thickness: 0.6pt, dash: "dashed"))
  ctz-draw(segment: ("B", "Mb"), stroke: (paint: purple, thickness: 0.6pt, dash: "dashed"))
  ctz-draw(segment: ("C", "Mc"), stroke: (paint: purple, thickness: 0.6pt, dash: "dashed"))

  ctz-draw(points: ("A", "B", "C", "G"), labels: (
    A: "left",
    B: "right",
    C: "above",
    G: "below"
  ))
})
```

# Perpendicular Bisectors and Circumcircle




```
#ctz-canvas(clip-canvas: (-1, -1, 7, 6), {
  import cetz.draw: *
  ctz-init()
  ctz-style(point: (shape: "dot", size: 0.08, fill: black))

  // Define triangle
  ctz-def-points("A", (0, 0), "B", (6, 0), "C", (2.5, 5))

  // Calculate midpoints
  ctz-def-midpoint("Mab", "A", "B")
  ctz-def-midpoint("Mbc", "B", "C")
  ctz-def-midpoint("Mca", "C", "A")

  // Circumcenter is intersection of perpendicular bisectors
  ctz-def-circumcenter("O", "A", "B", "C")

  // Create perpendicular bisector lines
  ctz-def-mediator("Pab1", "Pab2", "A", "B")
  ctz-def-mediator("Pbc1", "Pbc2", "B", "C")
  ctz-def-mediator("Pca1", "Pca2", "C", "A")

  // Draw triangle
  ctz-draw(line: ("A", "B", "C", "A"), stroke: black + 1.5pt)

  // Draw perpendicular bisectors (clipped)
  ctz-draw-segment("Pab1", "Pab2", stroke: blue + 0.8pt)
  ctz-draw-segment("Pbc1", "Pbc2", stroke: red + 0.8pt)
  ctz-draw-segment("Pca1", "Pca2", stroke: green + 0.8pt)

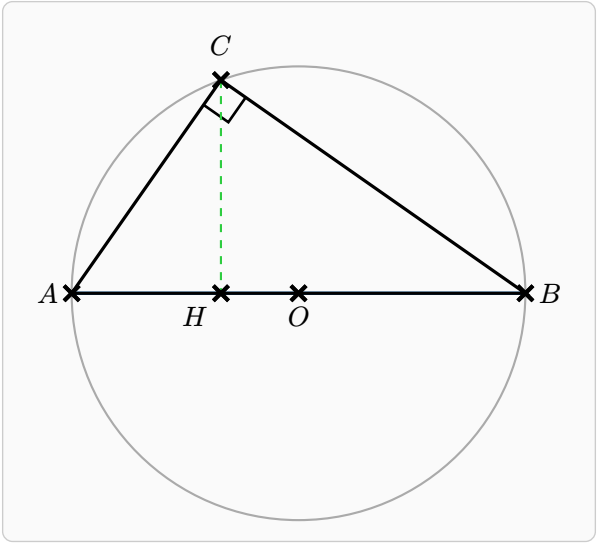
  // Draw circumcircle
  ctz-draw(circumcircle: ("A", "B", "C"), stroke: purple + 1.2pt)

  // Radii (equal length)
  ctz-draw(segment: ("O", "A"), stroke: gray + 0.4pt)
  ctz-draw(segment: ("O", "B"), stroke: gray + 0.4pt)
  ctz-draw(segment: ("O", "C"), stroke: gray + 0.4pt)

  ctz-draw-mark-right-angle("A", "Mab", "O", size: 0.3)
  ctz-draw-mark-right-angle("B", "Mbc", "O", size: 0.3)
  ctz-draw-mark-right-angle("C", "Mca", "O", size: 0.3)

  ctz-draw(points: ("A", "B", "C", "O", "Mab", "Mbc", "Mca"), labels: (
    A: "left",
    B: "right",
    C: "above",
    O: "below"
  ))
})
```

# Thales' Theorem




```
#ctz-canvas({
  import cetz.draw: *
  ctz-init()
  ctz-style(point: (shape: "cross", size: 0.1, stroke: black + 1.5pt))

  // Circle with diameter AB
  ctz-def-points("O", (0, 0), "A", (-3, 0), "B", (3, 0))

  // Point C on circle
  ctz-def-point-on-circle("C", "O", 3, 110)

  // Draw circle and diameter
  ctz-draw(circle-r: ("O", 3), stroke: gray + 0.8pt)
  ctz-draw(segment: ("A", "B"), stroke: blue + 1.2pt)

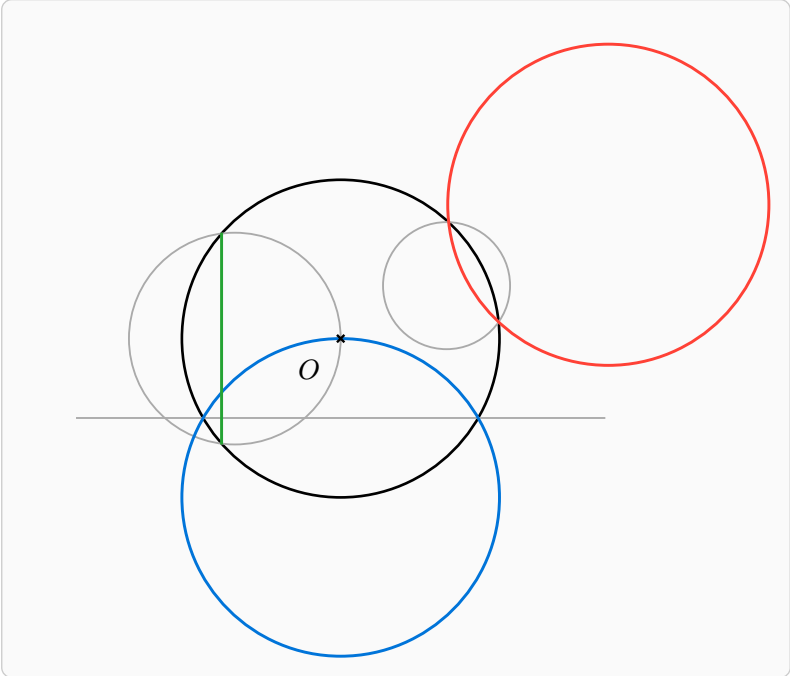
  // Draw triangle ACB
  // By Thales' theorem: angle ACB = 90° (inscribed in semicircle)
  ctz-draw(line: ("A", "C", "B", "A"), stroke: black + 1.2pt)

  // Mark the right angle at C
  ctz-draw-mark-right-angle("A", "C", "B", size: 0.4)

  // Draw altitude from C to AB
  ctz-def-project("H", "C", "A", "B")
  ctz-draw(segment: ("C", "H"), stroke: (paint: green, thickness: 0.8pt, dash: "dashed"))

  ctz-draw(points: ("A", "B", "C", "O", "H"), labels: (
    A: "left",
    B: "right",
    C: "above",
    O: "below",
    H: "below left"
  ))
})
```

# Inversion




```
#ctz-canvas(clip-canvas: (-6, -5, 6, 6), {
  import cetz.draw: *
  ctz-init()

  ctz-def-points(
    O: (0, 0),
    A: (-5, -1.5),
    B: (5, -1.5),
    C: (2, 1),
    D: (-2, 0),
  )

  ctz-def-line("l1", "A", "B")
  ctz-def-circle("c1", "C", radius: 1.2)
  ctz-def-circle("c2", "D", radius: 2)

  ctz-def-inversion("l1i", "l1", "O", 3)
  ctz-def-inversion("c1i", "c1", "O", 3)
  ctz-def-inversion("c2i", "c2", "O", 3)

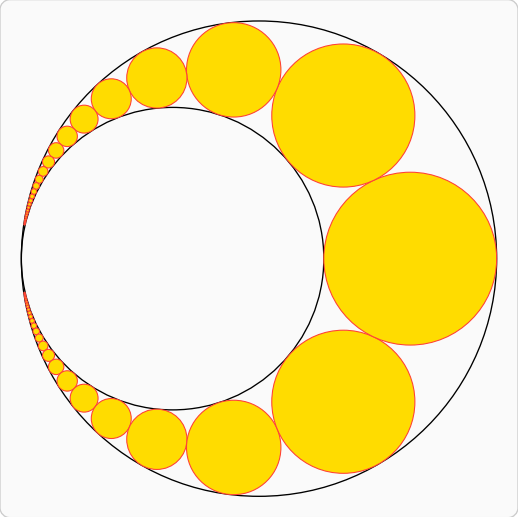
  ctz-draw(circle-r: ("O", 3), stroke: black + 1pt)

  ctz-draw("l1", stroke: gray + 0.7pt)
  ctz-draw("c1", stroke: gray + 0.7pt)
  ctz-draw("c2", stroke: gray + 0.7pt)

  ctz-draw("l1i", stroke: blue + 1.1pt)
  ctz-draw("c1i", stroke: red + 1.1pt)
  ctz-draw("c2i", stroke: green.darken(20%) + 1.1pt)

  ctz-draw(points: ("O"), labels: (O: "below left"))
})
```

# Inversion Packing



```
#ctz-canvas({
  import cetz.draw: *
  ctz-init()

  let n = 24

  // Inversion circle setup
  ctz-def-points(O: (-4.5, 0))
  ctz-def-line("l1", (-1, 0), (-1, 1))
  ctz-def-line("l2", (1, 0), (1, 1))

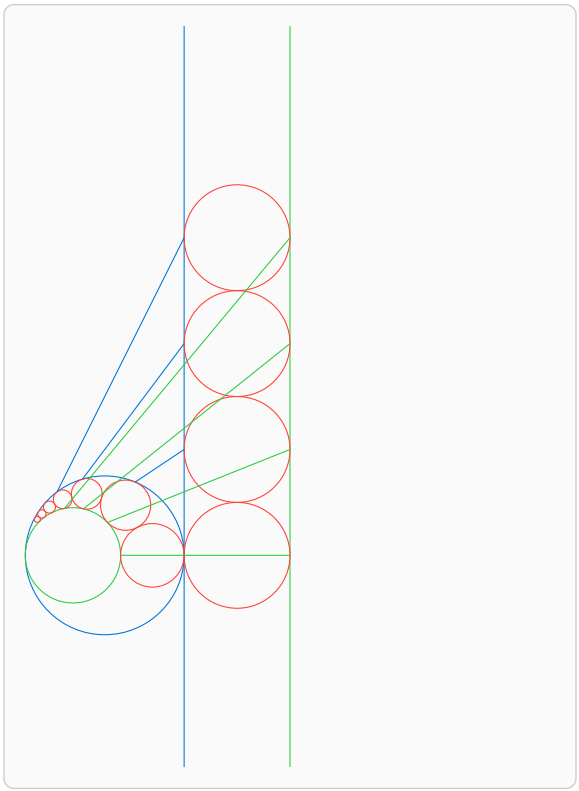
  ctz-def-inversion("l1i", "l1", "O", 1)
  ctz-def-inversion("l2i", "l2", "O", 1)

  // Draw inverted lines (outer boundary)
  ctz-draw("l1i", stroke: black + 0.5pt)
  ctz-draw("l2i", stroke: black + 0.5pt)

  // Invert a stack of circles to create the packing
  for i in range(-n, n + 1) {
    let name = "c" + str(i)
    let invname = "ci" + str(i)
    ctz-def-circle(name, (0, 2 * i), radius: 1)
    ctz-def-inversion(invname, name, "O", 1)

    ctz-draw(invname, fill: yellow, stroke: red + 0.4pt)
  }
})
```

# Inversion Ladder



```
#ctz-canvas(clip-canvas: (-4, -4, 6, 10), {
  import cetz.draw: *
  ctz-init()

  let O = (-4, 0)
  let R = 3
  let n = 6

  ctz-def-line("l1", (-1, 0), (-1, 1))
  ctz-def-line("l2", (1, 0), (1, 1))
  ctz-def-inversion("l1i", "l1", O, R)
  ctz-def-inversion("l2i", "l2", O, R)

  ctz-draw-line-add((-1, 0), (-1, 1), add: (10, 10), stroke: blue + 0.4pt)
  ctz-draw-line-add((1, 0), (1, 1), add: (10, 10), stroke: green + 0.4pt)
  ctz-draw("l1i", stroke: blue + 0.4pt)
  ctz-draw("l2i", stroke: green + 0.4pt)

  for i in range(0, n + 1) {
    let cname = "C" + str(i)
    let cpname = "Cp" + str(i)
    ctz-def-circle(cname, (0, 2 * i), radius: 1)
    ctz-def-inversion(cpname, cname, O, R)
    ctz-draw(cpname, stroke: red + 0.4pt)

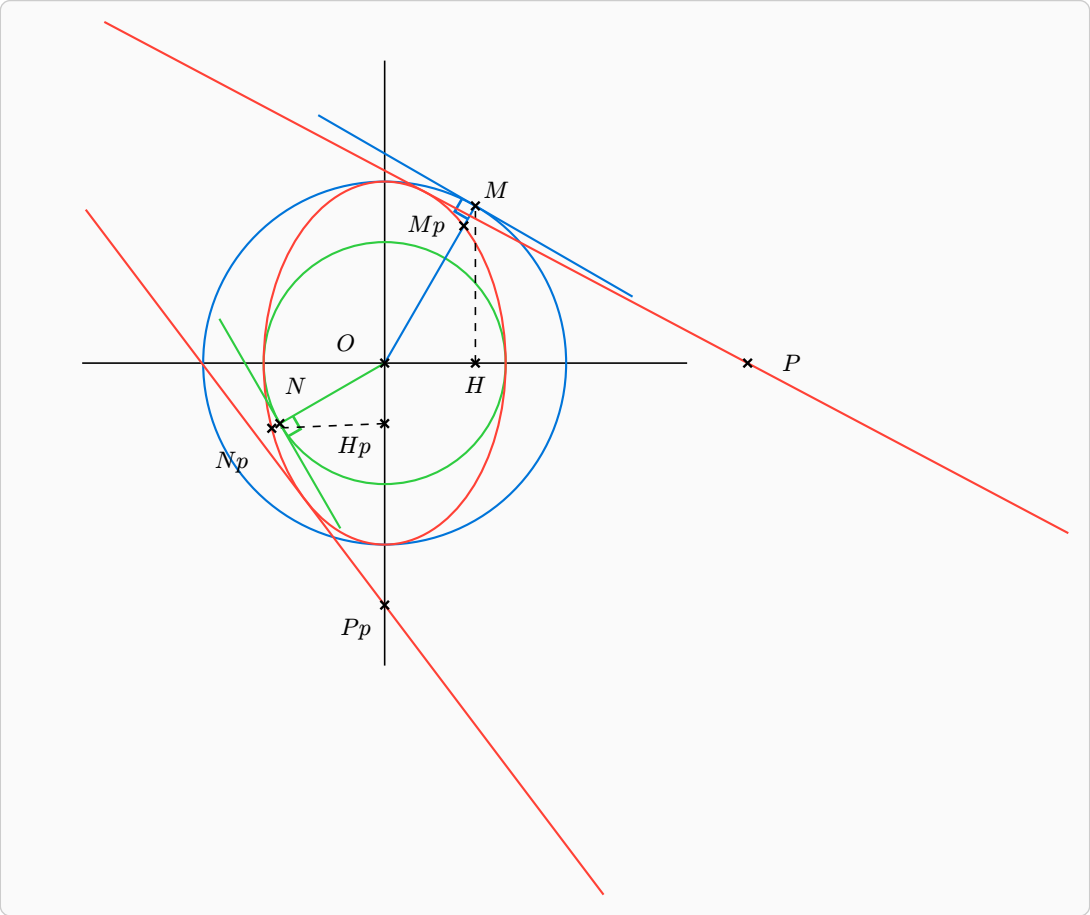
    if calc.abs(i) < 4 {
      ctz-draw(cname, stroke: red + 0.4pt)

      ctz-def-points(P1: (1, 2 * i), P2: (-1, 2 * i))
      ctz-def-inversion("P1i", "P1", O, R)
      ctz-def-inversion("P2i", "P2", O, R)

      ctz-draw(segment: ("P1", "P1i"), stroke: green + 0.4pt)
      ctz-draw(segment: ("P2", "P2i"), stroke: blue + 0.4pt)
    }
  }
})
```

# 15. Advanced Examples

### Ellipse and Circles Tangency




```
#ctz-canvas(length: 0.8cm, {
  import cetz.draw: *
  ctz-init()

  let a = 3
  let b = 2

  let blue-stroke = (paint: blue, thickness: 0.8pt)
  let green-stroke = (paint: green, thickness: 0.8pt)
  let red-stroke = (paint: red, thickness: 0.8pt)

  // Axes
  ctz-def-points(O: (0, 0), X1: (-5, 0), X2: (5, 0), Y1: (0, -5), Y2: (0, 5))
  ctz-draw(line: ("X1", "X2"), stroke: (paint: black, thickness: 0.6pt))
  ctz-draw(line: ("Y1", "Y2"), stroke: (paint: black, thickness: 0.6pt))

  // Circles and ellipse
  ctz-def-circle("C", "O", radius: a)
  ctz-def-circle("Cp", "O", radius: b)
  ctz-draw("C", stroke: blue-stroke)
  ctz-draw("Cp", stroke: green-stroke)
  ctz-draw(ellipse: ("O", b, a, 0deg), stroke: red-stroke)

  // Key points H, H'
  ctz-def-points(H: (a / 2, 0), Hp: (0, -b / 2))

  // Lines through H and H'
  ctz-def-perp("L1a", "L1b", ("X1", "X2"), "H")
  ctz-def-para("L2a", "L2b", ("X1", "X2"), "Hp")
```

```
  // Intersections with circles
  ctz-def-lc(("M", "M2"), ("L1a", "L1b"), "C", near: "H")
  ctz-def-lc(("N", "N2"), ("L2a", "L2b"), "Cp", near: "Hp")

  // Project M, N onto ellipse
  ctz-def-ellipse-project("Mp", "O", b, a, "M")
  ctz-def-ellipse-project("Np", "O", b, a, "N")

  // Dashed construction
  ctz-draw(line: ("H", "M"), stroke: (paint: black, thickness: 0.6pt, dash: "dashed"))
  ctz-draw(line: ("Hp", "Np"), stroke: (paint: black, thickness: 0.6pt, dash: "dashed"))

  // Radii
  ctz-draw(line: ("O", "M"), stroke: blue-stroke)
  ctz-draw(line: ("O", "N"), stroke: green-stroke)

  // Tangents at M and N
  ctz-def-tangent-at("TgM1", "TgM2", "M", "O")
  ctz-def-tangent-at("TgN1", "TgN2", "N", "O")
  ctz-draw(line: ("TgM1", "TgM2"), stroke: blue-stroke)
  ctz-draw(line: ("TgN1", "TgN2"), stroke: green-stroke)

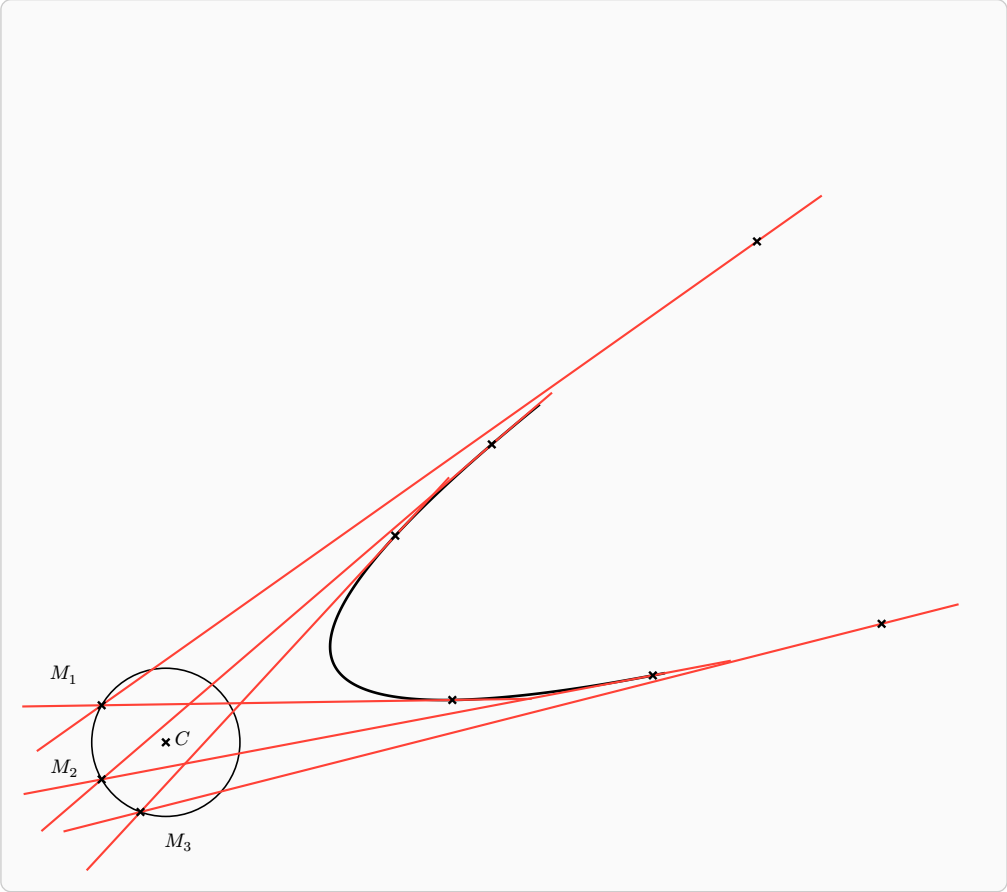
  // Intersections with axes
  ctz-def-ll("P", ("TgM1", "TgM2"), ("X1", "X2"))
  ctz-def-ll("Pp", ("TgN1", "TgN2"), ("Y1", "Y2"))

  // Perpendicular marks
  ctz-draw-mark-right-angle("TgM1", "M", "O", color: blue)
  ctz-draw-mark-right-angle("TgN1", "N", "O", color: green)

  // Ellipse tangents from P, P'
  ctz-def-ellipse-tangent-from("Tp1", "Tp2", "O", b, a, "P", "Mp")
  ctz-def-ellipse-tangent-from("Tpp1", "Tpp2", "O", b, a, "Pp", "Np")
  ctz-draw(line: ("Tp1", "Tp2"), stroke: red-stroke, mark: (end: ">", start: ">"))
  ctz-draw(line: ("Tpp1", "Tpp2"), stroke: red-stroke, mark: (end: ">", start: ">"))

  // Labels
  ctz-draw(points: ("O", "H", "Hp", "M", "Mp", "N", "Np", "P", "Pp"), labels: (
    O: (pos: "below left", offset: (-0.15, -0.15)),
    H: "below", Hp: "left", M: "above right", Mp: "above",
    N: "left", Np: "below left", P: "right", Pp: "below"
  ))
})
```

# Parabola Tangents from Circle




```
#ctz-canvas(length: 0.7cm, clip-canvas: (-4, -4, 12, 12), {
  import cetz.draw: *
  import "@preview/ctz-euclide:0.2.1": *
  ctz-init()

  let F = (0, 0)
  let p = 0.35
  let theta = 25deg

  // Draw parabola
  ctz-draw-parabola-focus(F, p, angle: theta, extent: 2.8,
    stroke: (paint: black, thickness: 1pt))

  let (D1, D2, V) = parabola-directrix-raw(F, p, angle: theta)

  // Circle positioned to be tangent to parabola
  let C = (V.at(0) + 10 * (V.at(0) - F.at(0)),
           V.at(1) + 10 * (V.at(1) - F.at(1)))
  let r = 1.4
  ctz-draw(circle-r: (C, r), stroke: (paint: black, thickness: 0.6pt))

  // Three points on circle
  let M1 = (C.at(0) + r * calc.cos(150deg), C.at(1) + r * calc.sin(150deg))
  let M2 = (C.at(0) + r * calc.cos(210deg), C.at(1) + r * calc.sin(210deg))
  let M3 = (C.at(0) + r * calc.cos(250deg), C.at(1) + r * calc.sin(250deg))

  // Draw tangent lines from each M point to parabola
  for M in (M1, M2, M3) {
    let tangents = parabola-tangents-from-point-raw(
      F, D1, D2, M, length: 1.5)
    for t in tangents {
      let (p1, p2, tp) = t
      ctz-draw(line: (p1, p2), stroke: (paint: red, thickness: 0.8pt))
      ctz-draw(points: (tp,), style: (shape: "dot", size: 2pt, fill: blue))
```

```
    }
  }

  // Label points
  ctz-def-points(C: C, M1: M1, M2: M2, M3: M3)
  ctz-draw(points: ("C", "M1", "M2", "M3"), labels: (
    C: (pos: "below", offset: (0, -0.25)),
    M1: (pos: "above left", offset: (-0.2, 0.15)),
    M2: (pos: "left", offset: (-0.2, 0)),
    M3: (pos: "below right", offset: (0.2, -0.15)),
  ))
})
```